مكتبة انوار الرافدين

معالجة اشارة

jafar walid

الكورس الاول

ثالث. هندسة تقنيات

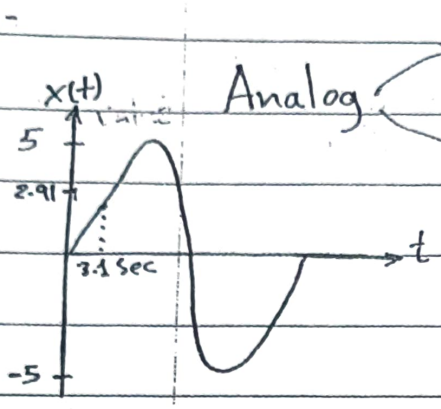
أ. ابراهيم حاتم

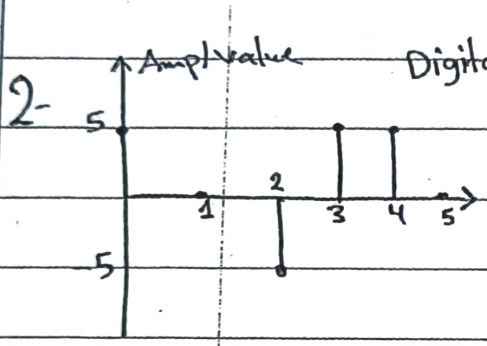
المكتبة الرئيسية والمعتمدة لكلية الرافدين هنا تجدون النسخة الأصلية
نتمنى لكم عام دراسياً موفقاً2025|/2026

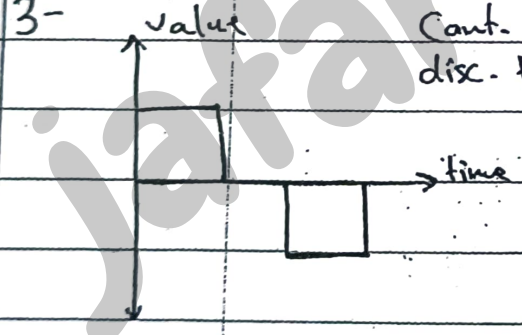
2016/10/16

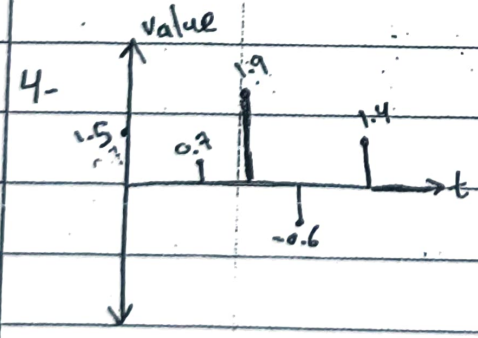
DSP

1-









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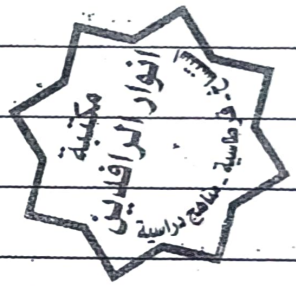
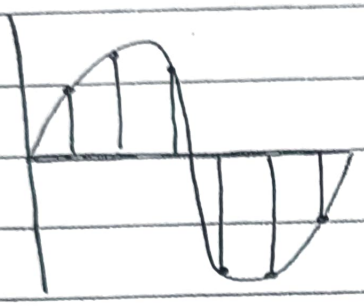
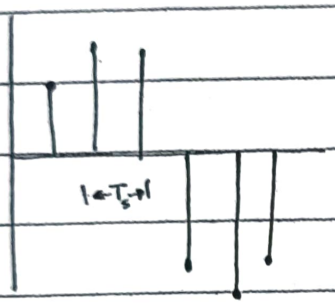


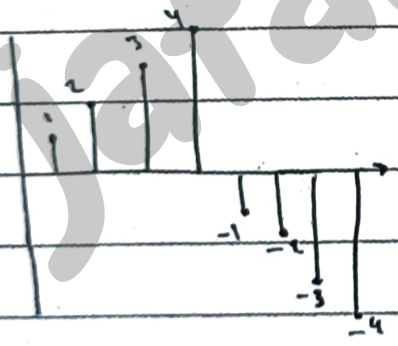
Sampling:

cont → disc
time time

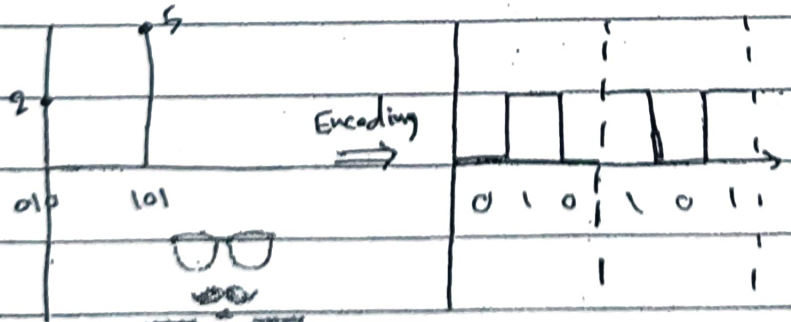




$T_s = 1/f_s$

Encoding:

8 levels

000	1
001	2
010	3
011	4
100	-1
101	-2
110	-3
111	-4



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Discrete time System :-

1. linear or non-linear.
2. Time invariant or not.
3. Stabe or unstable.
4. Causal or non-Causal.

Ex: $y(n) = n\,x(n)$, check the system if it is ① linear ② Time invariant ③ stable ④ Causal

Solution:

① let i/p1 $x_1(n)$ — o/p1 $y_1(n)$

i/p2 $x_2(n)$ — o/p2 $y_2(n)$

i/p3 $x_1(n) + x_2(n)$ — o/p3 $y_3(n)$

$\therefore y_1(n) = n\,x_1(n)$

$y_2(n) = n\,x_2(n)$

$y_1(n) + y_2(n) = n\,x_1(n) + n\,x_2(n)$

for i/p3

$y_3(n) = n\,[x_1(n) + x_2(n)]$

$= n\,x_1(n) + n\,x_2(n)$

if $y_3(n) \overset{?}{=} y_1(n) + y_2(n)$

$=$ linear ✓

$\neq$ non-linear

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② let ① $n \longrightarrow (n-n_0)$: خاصية ثابتة

$y_1(n-n_0) = (n-n_0)[x(n-n_0)]$

let ② $x(n) \longrightarrow x(n-n_0)$

$y_2(n) = n\,x(n-n_0)$

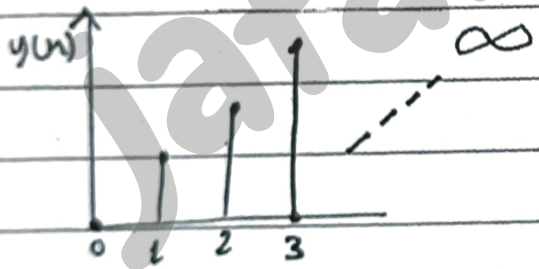
if $y_1(n-n_0) = y_2(n)$ Time invariant لا يتغير مع الزمن

if $y_1(n-n_0) \neq y_2(n)$ Time varying يتغير مع الزمن

$\therefore y_1(n-n_0) \neq y_2(n) \therefore$ Time varying

③ let $i/p = u(n)$ bounded i/p

$y(n) = n\,u(n)$ (ramp) unbounded $\therefore$ unstable



impulse response

if $S = \sum_{n=-\infty}^{\infty} |h(n)|$

if $S < \infty$ Stable

$y(n) = n\,x(n)$ difference equation

$h(n) = y(n)\big]_{x(n)=\delta(n)} = n\,\delta(n)$

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Ex/ find linear convolution of $x_1(n)$ & $x_2(n)$

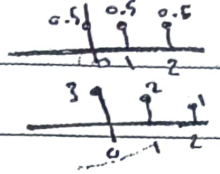
*linear Convolution *

*Circular Convolution (*)

$$y(n) = x_1(n) * x_2(n)$$

$x_1(n) = [0.5 \overset{n=0}{\leftarrow} \ 0.5 \quad 0.5]$

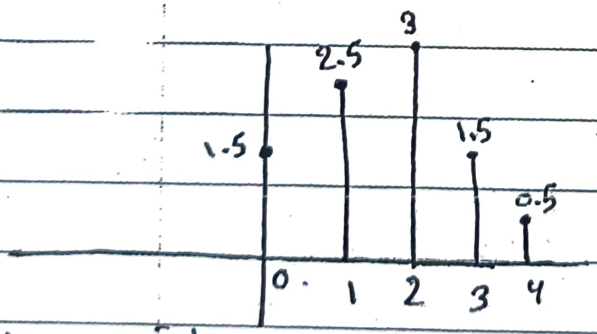
$x_2(n) = [3 \overset{n=0}{\leftarrow} \ 2 \quad 1]$



Sol/ using Tabular Method

x_2 \ x_1	n=0 → 0.5	0.5	0.5
n=0 → 3	1.5	1.5	1.5
2	1	1	1
1	0.5	0.5	0.5

$$y(n) = [1.5, 2.5, 3, 1.5, 0.5]$$



عدد عينات الاشارة الثانية — عدد عينات الاشارة الاولى

$$l = N_1 + N_2 - 1$$

$$l = 3 + 3 - 1 = 5$$

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using graphical method

X_1	0.5	0.5	0.5
X_2	3	2	1

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$\therefore h(n) = n\,\delta(n)$

$\delta(0) = 1$

$\delta(-1) = 0$

$\delta(+1) = 0$

⋮

④ System is Causal if it does not depend on future i/p.

$X(n)$ current Present i/p

$X(n+n_0)$ future i/p

$X(n-n_0)$ Past i/p

$y(n) = n\,X(n)$ Causal

$n=0$ $y(0) = 0$

$n=1$ $y(1) = X(1)$

$y(2) = 2\,X(2)$

$2 \geq 2$ Causal

Ex: $y(n) = n\,X(n+2)$, causal or not?

Solution:

$n=0$ $y(0) = 0$

$y(1) = X(3)$

$1 < 3$ non-causal

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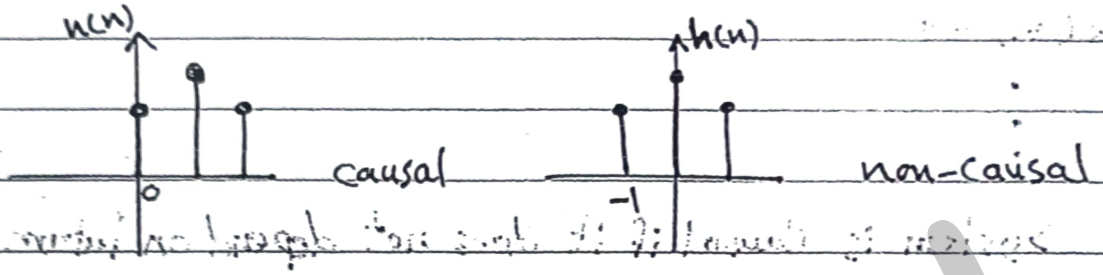
another Solution:

if $h(n)=0$

for $n<0$



Ex: $y(n) = e^{x(n)}$, check if it is Stable

$e^{u(n)}$

$n=0 \quad y = e^{u(0)} = e^{1} = 2.7$

$n=1 \quad y = e^{u(1)} = e^{0} = 2.7$

bounded o/p (Stable)

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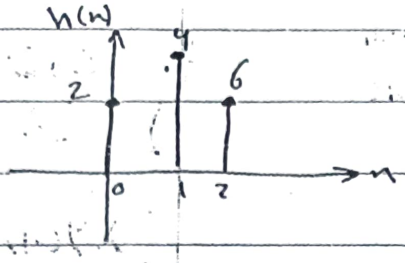
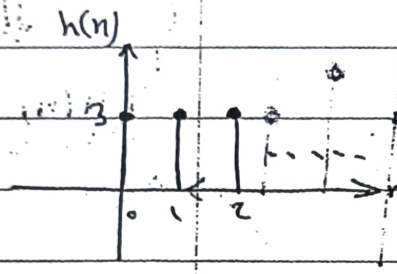
Finite impulse response (FIR) and Infinite Impulse Response (IIR)

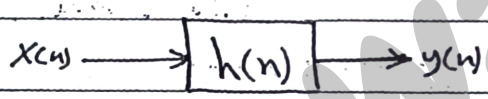


← FIR

h(n): impulse response



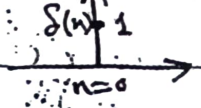
← IIR



P41 Ex: Solve the D.E [y(n)=?] assume y(n)=0 for n<0

$x(n) = \delta(n)$

$y(n) - a\,y(n-1) = x(n)$ D.E



$\delta(0) = 1$

$\delta(-1) = 0$

$\delta(1) = 0$

Solution:

$y(n) - a\,y(n-1) = \delta(n)$

at $n=0$ $\quad y(0) - a\,y(-1) = \delta(0) \Rightarrow y(0) = 1$

at $n=1$ $\quad y(1) - a\,y(0) = \delta(1) \Rightarrow y(1) = a$

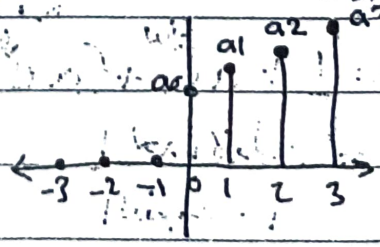
at $n=2$ $\quad y(2) - a\,y(1) = \delta(2) \Rightarrow y(2) = a^2$

at $n=3$ $\quad y(3) - a\,y(2) = \delta(3) \Rightarrow y(3) = a^3$

$\therefore y(n) = a^n \quad n \geq 0$

$$y(n) = \begin{cases} 0 & n<0 \\ a^n & n \geq 0 \end{cases}$$

or $y(n) = a^n u(n)$



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1.3.5

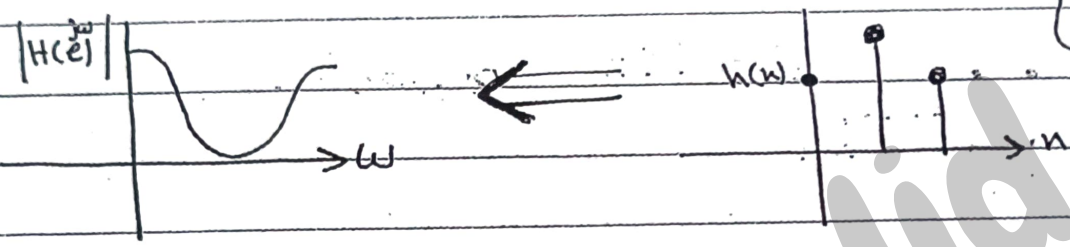
Freq. domain Representations

$$H(e^{jw}) = \sum_{k=-\infty}^{\infty} h(k)\, e^{-jwk}$$

Fourier Transform of discrete time Signal

$H(e^{jw})$: Frequency Response; $h(k)$: Impulse Response

$y(n) = a\,x(n) \Rightarrow h(n) = a\,\delta(n)$



Frequency domain — Time Domain

$$H(e^{jw}) = H_R(e^{jw}) + j\,H_I(e^{jw}) = |H(e^{jw})|\,\angle H(e^{jw})$$

$$= r\,\angle\theta$$

$$= |H(e^{jw})|\exp\{j\arg[H(e^{jw})]\}$$

$$= re^{j\theta}$$

مثلاً $5e^{j\frac{\pi}{2}} = 5\angle 90^\circ$

p46 Ex: Find Freq. Response given $h(n) = a^n u(n)$, $|a| < 1$

Solution:

$$H(e^{jw}) = \sum_{n=-\infty}^{\infty} a^n u(n)\, e^{-jwn}$$

$$= \sum_{n=0}^{\infty} a^n e^{-jwn} = \sum_{n=0}^{\infty} (ae^{-jw})^n$$

$$= 1 + ae^{-jw} + (ae^{-jw})^2 + \cdots$$

$$= \frac{\text{الحد الأول}}{1 - \text{النسبة}}$$

النسبة $= \frac{ae^{-jw}}{1} = \frac{(ae^{-jw})^2}{ae^{-jw}} = ae^{-jw}$

بشرط $|ae^{-jw}| < 1$

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$$\therefore H(e^{jw}) = \frac{1}{1 - ae^{-jw}}$$

$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$

$$\therefore H(e^{jw}) = \frac{1}{1 - a[\cos w - j\sin w]} = \frac{1}{1 - a\cos w - j(a\sin w)}$$

$$|H(e^{jw})| = \frac{|1|}{\text{(مقام)}} = \frac{1}{\sqrt{(1 - a\cos w)^2 + (a\sin w)^2}}$$

$$\angle H(e^{jw}) = \angle \text{البسط} - \angle \text{المقام} = 0 - \left[\tan^{-1}\left(\frac{\pm a\sin w}{1 - a\cos w}\right)\right]$$

(تخيلي / حقيقي)

Ex: Find the following if $h(n) = \frac{1}{2}\delta(n) + \delta(n-1) + \frac{1}{2}\delta(n-2)$

① $H(e^{jw})$, $|H(e^{jw})|$, $\angle H(e^{jw})$

Solution:

$$H(e^{jw}) = \sum_{n=-\infty}^{\infty} h(n)\, e^{-jwn}$$

$$= \sum_{n=-\infty}^{\infty} \left[\frac{1}{2}\delta(n) + \delta(n-1) + \frac{1}{2}\delta(n-2)\right] e^{-jwn}$$

$$= \sum \frac{1}{2}\delta(n)e^{-jwn} + \sum \delta(n-1)e^{-jwn} + \sum \frac{1}{2}\delta(n-2)e^{-jwn}$$

$$= \cdots 0 + 0 + \frac{1}{2}\delta(0)e^{-jw0} + 0 + 0 \cdots + \delta(1-1)e^{-jw1} + \frac{1}{2}\delta(0)e^{-jw2}$$

($\delta(0) \to 1$, $\delta(1-1) \to 1$, $\delta(0) \to 1$, $\delta(2-2)$)

$$H(e^{jw}) = \frac{1}{2} + e^{-jw} + \frac{1}{2}e^{-2jw}$$

$$= e^{-jw}\left[\frac{1}{2}e^{+jw} + 1 + \frac{1}{2}e^{-jw}\right]$$

$$* \cos w = \frac{e^{jw} + e^{-jw}}{2}$$

$$H(e^{jw}) = e^{-jw}[1 + \cos w]$$

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$\therefore |H(e^{jw})| = [1 + \cos w]$ $\quad H(e^{jw}) = |H| \angle H$:

$\angle H(e^{jw}) = -w$ $\quad = r \angle \theta$

$= r\, e^{j\theta}$

Magnitude, Phase

② $y_{ss}(n)$, if $X(n) = \overset{A}{5} \cos\left(\overset{w_o}{\frac{\pi}{4}} n\right) + \overset{\phi}{0}$

Solution:

IP $y_{ss}(n) = A * |H(e^{jw_o})| \cos[w_o n + \phi + \angle H(e^{jw_o})]$ ثابت ضرب

IP if $X(n) = A \cos[w_o n + \phi]$ ثابت جمع

$$X(n)_{w_o} \longrightarrow |H(e^{jw_o})| \longrightarrow y_{ss}(n)_{w_o}$$

* الحاسبة توضع على الـ Rad

$A = 5, \; w_o = \frac{\pi}{4}, \; \phi = 0$

$\therefore |H| = \left[\frac{1 + \cos\frac{\pi}{4}}{}\right] = 1 + \frac{1}{\sqrt{2}} = 0.707$

$\angle H = -\frac{\pi}{4}$

$\therefore y_{ss}(n) = 5 * 1.707 \cos\left[\frac{\pi}{4} n + 0 + \left(-\frac{\pi}{4}\right)\right]$ LPF

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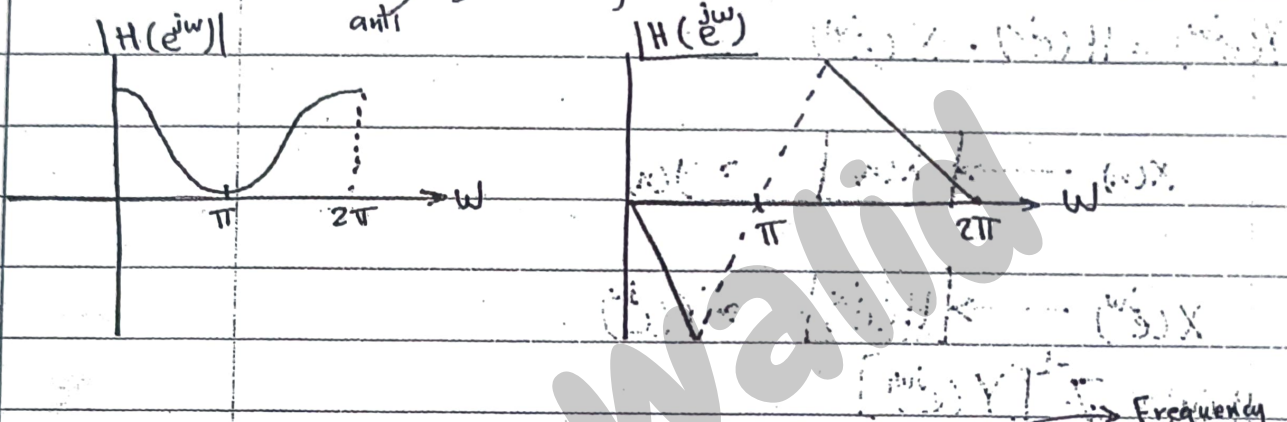
Properties of freq. response: $H(e^{jw})$

1- w is Continous.

2- $H(e^{jw})$ is periodic with period 2π

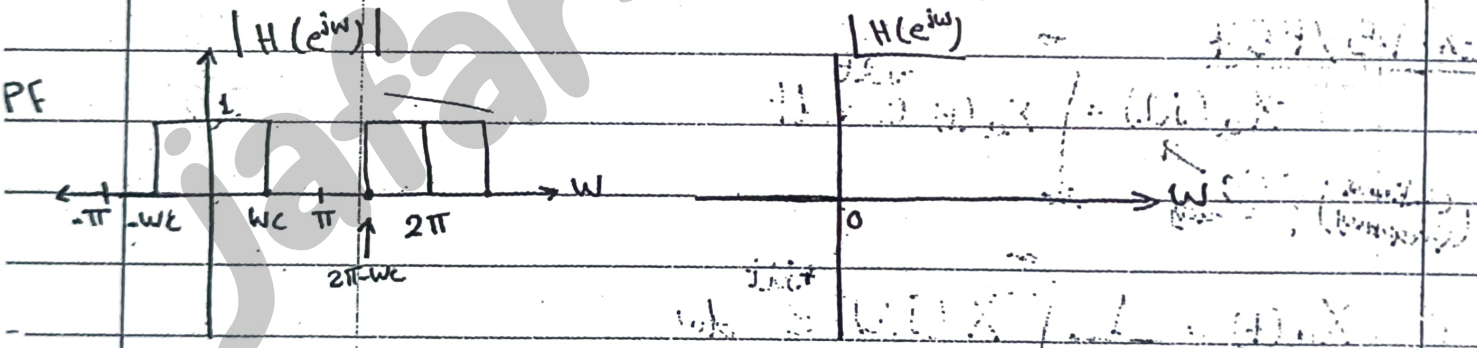
3- $|H(e^{jw})|$ is even Symmetry about π

4- $|H(e^{jw})|$ is odd Symmetry about π (anti)



Frequency Respanse

Ex: Find impulse response h(n) for givin $H(e^{jw})$



Solution: ظرفة ثابت

$$h(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(e^{jw})\, e^{jwn}\, dw$$

inverse fourier transform (discrete time Signal)

$$h(n) = \frac{1}{2\pi}\int_{-wc}^{wc} 1 \cdot e^{jwn}\, dw = \frac{1}{2\pi}\left[\frac{e^{jwn}}{jn}\right]_{-wc}^{wc}$$

$$= \frac{1}{2\pi}\left[\frac{e^{jwcn} - e^{-jwcn}}{jn}\right] = \frac{1}{\pi n}\,\frac{\sin(wcn)}{2fc} * 2fc$$

$$= 2fc\, \mathrm{Sinc}[wcn]$$

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Energy:

$$E = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{jw})\,X^*(e^{jw})\,dw \quad \text{(freq. domain)}$$

(complex conjugate; discrete frequency)

$$= \sum_{n=-\infty}^{\infty} x(n)\,x^*(n) \quad \text{(time domain)}$$

$$Y(e^{jw}) = H(e^{jw}) \cdot X(e^{jw})$$

$x(n) \rightarrow [\,h(n)\,] \rightarrow y(n)$

$X(e^{jw}) \rightarrow [\,H(e^{jw})\,] \rightarrow Y(e^{jw})$

$y(n) = \mathcal{Z}^{-1}[Y(e^{jw})]$

Ex: 1.5/P57

$$X_a(j\Omega) = \int_{-\infty}^{\infty} x_a(t)\,e^{-j\Omega t}\,dt$$

(Continuous Frequency) Analog Frequency

$$\Rightarrow x_a(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\Omega)\,e^{+j\Omega t}\,dw$$

Fourier Transform Pair For Cont. time Signal

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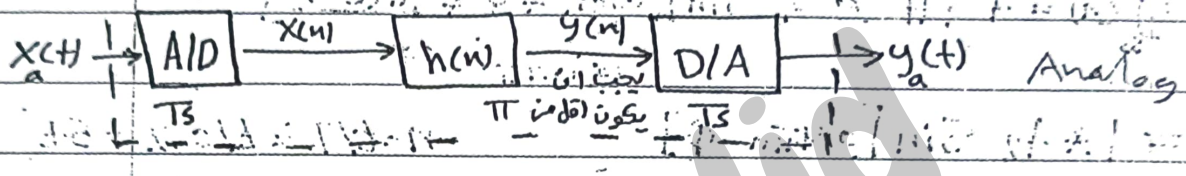
Digital Filter with A/D and D/A

$x(n) \rightarrow \boxed{h(n)} \rightarrow y(n) = x(n) * h(n)$ (linear convolution)

all in discrete time — Signal, System



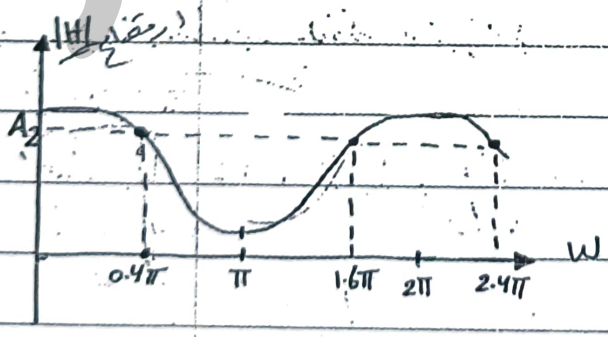
① equivalent analog filter

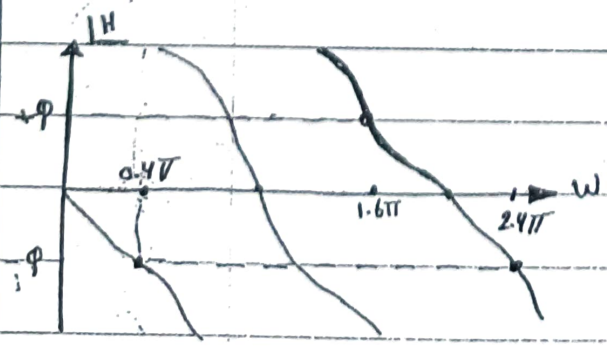
② A/D - H(z) - D/A structure

$$\omega = \Omega T$$

ω: digital frequency (rad), Ω: analog frequency (rad/s)

Ex: if $x_1(t) = \sin(2\pi t)$, $x_2(t) = \sin(12\pi t)$, find $y_{ss1}(t)$, $y_{ss2}(t)$ for given $H(e^{j\omega})$, if $T = 0.2$ Sec in A/D-H(z)-D/A structure





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Solution:

① $x_1(t) \rightarrow x_1(n)$

$x_1(n) = \sin(2\pi n T) = \sin(2\pi n * 0.2)$

$x_1(n) = \sin(0.4\pi n)$

$\Omega_1 = 2\pi$ rad/sec $\rightarrow w_1 = \Omega_1 T = 2\pi * 0.2 = 0.4\pi$ rad

$y_{ss}(n) = A\,|H(e^{jw})|\,\sin[w_0 n + \Phi + |H(e^{jw_0})|]$

($w_0 = 0.4\pi$ ، $\Phi = 0$)

$= 1 * A_2 \sin[0.4\pi n - \Phi]$ $\quad n = t/T = t/0.2 = 5t$

$\therefore y_{ss}(t) = A_2 \sin[2\pi t - \Phi]$

② $x_2(t) \rightarrow x_2(n)$

$x_2(n) = \sin(2.4\pi n)$

$\Omega_2 = 12\pi \rightarrow w_2 = \Omega_2 T = 12\pi * 0.2 = 2.4\pi$ (rad)

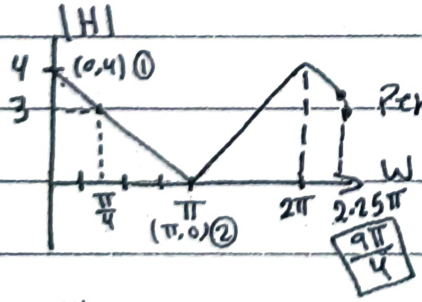
$y_{ss}(n) = 1 * A_2 \sin[0.4\pi n - \Phi]$

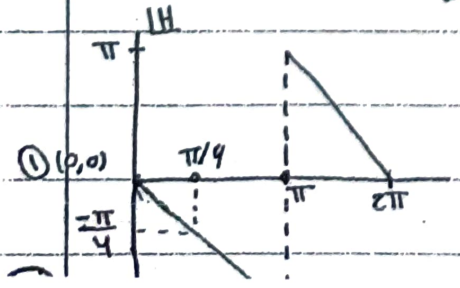
$y_{ss}(t) = A_2 \sin[2\pi t - \Phi]$ (صورة مشوهة للطابقات الناجين)

* للتخلص من التشوه نضع Analog Filter في بداية الـ System للتخلص من الإشارة أعلى من π

Ex: find $y_{ss}(n)$ for: ① $x(n) = 5\cos\left(\frac{\pi n}{4}\right)$, ② $x(n) = 5\cos\left(\frac{9\pi n}{4}\right)$

$x(t) \rightarrow x(n)$ $\quad w_0$





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Solution: ① $|H| = ?$ & $\angle H = ?$ $A = 5,\ \phi = 0,\ \omega_0 = \frac{\pi}{4}$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} \rightarrow \frac{y - 4}{x - 0} = \frac{0 - 4}{\pi - 0}$$

$$y - 4 = \frac{-4}{\pi} x$$

$$y - 4 = -1$$

for $\angle H$

$$\frac{y - 0}{x - \pi} = -1$$

$$y = -x$$

$$\angle H = \frac{-\pi}{4}$$

$$y_{ss}(n) = A|H| \cos[\omega_0 n + \phi + \angle H]$$

$$= 5 * 3 \cos\left[\frac{\pi}{4} n + 0 - \frac{\pi}{4}\right]$$

Solution: ② $|H| = ?$, $\angle H = ?$

$|H| = 3$

$\angle H = \frac{-\pi}{4}$

aliased freq.

$$\therefore y_{ss}(n) = 5 * 3 \cos\left[\frac{\pi}{4} n + 0 - \frac{\pi}{4}\right]$$

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Chapter.2.

Z-Transform

2016/11/2

$$Z\{X(n)\} = \sum_{-\infty}^{\infty} X(n) z^{-n} = X(z) \text{ with ROC}$$

(ROC: Region of convergence (radius))

Table 2.1:

1) $X(n) = \delta(n) \longrightarrow X(z) = 1$ with ROC: all z — impulse
2) $x(n) = u(n) \longrightarrow X(z) = \frac{z}{z-1}$ with ROC: $|z| > 1$ — unit step
3) $x(n) = -u(-n-1) \longrightarrow X(z) = \frac{z}{z-1}$ with ROC: $|z| < 1$ — unit step
4) $X(n) = a^n u(n) \longrightarrow X(z) = \frac{z}{z-a}$ with ROC: $|z| > a$ — exponential
5) $X(n) = -b^n u(-n-1) \longrightarrow X(z) = \frac{z}{z-b}$ with ROC: $|z| < b$ — exponential
6) $X(n) = n u(n) \longrightarrow X(z) = \frac{z}{(z-1)^2}$ with ROC: $|z| > 1$
7) $X(n) = \sin(w_0 n)\, u(n) \longrightarrow X(z) = \frac{z \sin w_0}{z^2 - 2z\cos w_0 + 1}$ with ROC: $|z| > 1$
8) $X(n) = \cos(w_0 n)\, u(n) \longrightarrow X(z) = \frac{z(z - \cos w_0)}{z^2 - 2z\cos w_0 + 1}$ with ROC: $|z| > 1$

Z-Transform Properties

1. linearity
2. Translation (shifting)

if $Z\{x(n)\} = X(z)$ ROC 1

$\therefore Z\{x(n-n_0)\} = X(z) \cdot z^{-n_0}$ same ROC 1

3. Multiplication by n

if $Z\{x(n)\} = X(z)$ with ROC_1

$\therefore Z\{n * x(n)\} = -z \frac{d}{dz}[X(z)]$ with same ROC_1

4. Multiplication by a^n

if $Z\{x(n)\} = X(z)$ with ROC_1 $|z| > 1$

$\therefore Z\{a^n x(n)\} = X(z)\Big|_{z=\frac{z}{a}}$ with $ROC_{new} = a * ROC_1$ $|z| > a$

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⑤ Convolution:

$m(n) = x(n) \circledast y(n)$ time → Convolution

$M(z) = X(z) \odot Y(z)$ z-domain → ضرب

⑥ Initial Value Theorem

$$x(0) = \lim_{z \to \infty} X(z)$$

⑦ Final Value Theorem

$$x(\infty) = \lim_{z \to 1} \left\{ \left(\frac{z-1}{z} \right) X(z) \right\}$$

Ex: $x(n) = (n-2)\, a^{(n-2)} \cos[w_0(n-2)]$ Find $X(z)$

Solution:

$$z^{-2} \left\{ -z \frac{d}{dz} \mathcal{Z}\{a^n \cos w_0 n\} \right\}$$

$$= z^{-2} \left[-z \frac{d}{dz} \frac{\frac{z}{a}\left(\frac{z}{a} - \cos w_0\right)}{\left(\frac{z}{a}\right)^2 - 2\frac{z}{a}(\cos w_0 + 1)} \right]$$

Ex 2.6-F: $x(n) = (n+0.5)\left(\frac{1}{3}\right)^n u(n)$ Find $X(z)$ & ROC

Solution:

$$x(n) = n\left(\frac{1}{3}\right)^n u(n) + 0.5\left(\frac{1}{3}\right)^n u(n)$$

$$\therefore \left(\frac{1}{3}\right)^n u(n) \longrightarrow X(z) = \frac{z}{z - \frac{1}{3}} \quad \text{ROC: } |z| > \frac{1}{3}$$

$$\therefore n\left(\frac{1}{3}\right)^n u(n) \longrightarrow -z \frac{d}{dz}\left[\frac{z}{z-\frac{1}{3}} \right] = \frac{[(z-\frac{1}{3})*1] - [z*1]}{(z-\frac{1}{3})^2}$$

$$= \frac{+\frac{1}{3} z}{(z-\frac{1}{3})^2} \quad \text{with ROC: } |z| > \frac{1}{3}$$

$$\therefore X_{\text{total}}(z) = \frac{\frac{1}{3} z}{(z-\frac{1}{3})^2} * 0.5 \frac{z}{z - 1/3} \quad \text{with ROC: } |z| > \frac{1}{3}$$

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Ex: Find $\mathcal{Z}[u(-n)]$

Solution:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

$$X(z) = \sum_{n=-\infty}^{\infty} u(-n)\, z^{-n} = z^3 + z^2 + z^1 + 1 + (u(-1) * z^{-1}) \quad (u(-1) = 0)$$

$$= 1 + z + z^2 + \dots \infty$$

$$X(z) = \frac{1}{1-z} \qquad |z| < 1$$

* difference between $u(n)$ and $u(-n)$

$u(n)$	$u(-n)$
$X(z) = \frac{z}{z-1}$	$X(z) = \frac{1}{1-z}$

$n \longrightarrow (-n)$

$z \longrightarrow z^{-1}$

$$X(z) = \frac{z^{-1}}{z^{-1}-1} * \frac{z}{z} = \frac{1}{1-z}$$

Ex: $\mathcal{Z}[-n\, u(-n)]$

Solution:

$\because \mathcal{Z}\{n\, u(n)\} \Rightarrow X(z) = \frac{z}{(z-1)^2}$

$\because n \longrightarrow -n$

$z \longrightarrow z^{-1}$

$\therefore X(z) = \frac{z^{-1}}{(z^{-1}-1)^2}$ مقلوب السؤال

Ex: $x(n) = e^{-3n}\, u(n-1)$

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16/12/17

Solution:

$(e^{-3})^{n-1+1}\, u(n-1)$

$= (e^{-3})^{n-1} (e^{-3})^{1}\, u(n-1)$

$= e^{-3} z^{-1} \mathcal{Z}[(e^{-3})^{n} u(n)]$

$= e^{-3} z^{-1} * \frac{z}{z-e^{-3}}$ with ROC: $|z| > e^{-3}$

H.W/ $(2)^{-n} \sin\left(\frac{-\pi}{4} n\right) u(-n)$

Inverse Z-Transform:

① Complex integral (Residue Method)

② Partial fraction

③ Long division

Ex: $X(z) = \frac{z^4 + z^2}{(z+1)(z+2)(z+3)}$ Roc: $1 < |z| < 2$, find $x(n)$. using Partial fraction.

Solution:

$\frac{X(z)}{z} = \frac{z^3 + z}{(z+1)(z+2)(z+3)}$

* درجة المقام مساوية لدرجة البسط
لذلك استخدام PF لتكون درجة المقام أكبر من درجة البسط.
* سنستخدم القسمة هنا.

$* (z+1)(z+2)(z+3) = z^3 + 5z^2 + 6z + z^2 + 5z + 6$

$= z^3 + 6z^2 + 11z + 6$

X نسحب z من البسط
وضعه تحت X(z)

$$\begin{array}{r|l} z^3 + 6z^2 + 11z + 6 & z^3 + z \\ \hline & 1 \\ \mp z^3 \mp 6z^2 \mp 11z \mp 6 & \\ \hline -6z^2 + 10z - 6 & \end{array}$$

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$$\frac{X(z)}{z} = 1 - \frac{6z^2+10z+6}{(z+1)(z+2)(z+3)}$$

*now we can use P.F

$$\frac{A}{(z+1)} + \frac{B}{(z+2)} + \frac{C}{(z+3)}$$

$$A = \lim_{z\to -1} \frac{6z^2+10z+6}{(z+2)(z+3)} = \frac{6-10+6}{1*2} = 1$$

$$B = \lim_{z\to -2} \frac{6z^2+10z+6}{(z+1)(z+3)} = \frac{24-20+6}{(-1)(+1)} = -10$$

$$C = \lim_{z\to -3} \frac{6z^2+10z+6}{(z+1)(z+2)} = \frac{54-30+6}{(-2)(-1)} = 15$$

$$\therefore \frac{X(z)}{z} = 1 - \left[\frac{1}{z+1} - \frac{10}{z+2} + \frac{15}{z+3}\right]$$

$$X(z) = z - \frac{z}{z+1} + \frac{10z}{z+2} - \frac{15z}{z+3}$$

$|z| > 1$ موجب

$|z| < 2$ سالب

زمن موجب $a^n u(n) \leftarrow \frac{z}{z-a}$

زمن سالب $-b^n u(-n-1) \leftarrow \frac{z}{z-b}$

$$X(z) = \delta(n+1) - (-1)^n u(n) + \left[+10(-2)^n u(-n-1)\right] - \left[15 \quad (-3)^n u(-n-1)\right]$$

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Ex: $X(z) = \dfrac{z^2+z}{(z-\frac{1}{2})(z-\frac{1}{4})}$ solve using P.F , $X(n)=?$

Solution:

$\dfrac{X(z)}{z} = \dfrac{z+1}{(z-\frac{1}{2})^3(z-\frac{1}{4})}$ بعد بسط أقل من المقام

$\dfrac{X(z)}{z} = \dfrac{A}{(z-\frac{1}{2})^3} + \dfrac{B}{(z-\frac{1}{2})^2} + \dfrac{C}{(z-\frac{1}{2})} + \dfrac{D}{(z-\frac{1}{4})}$

$X(z) = A\dfrac{z}{(z-\frac{1}{2})} + B\dfrac{z}{(z-\frac{1}{2})} + C\dfrac{z}{(z-\frac{1}{2})} + D\dfrac{z}{(z-\frac{1}{4})}$

$X(n) = A\, n(n-1)\left(\frac{1}{2}\right)^{n-2} + B\, n(n-1)\left(\frac{1}{2}\right)^{n-2}$

Ex: Find $X(n)$ [only the first 3 samples] for $X(z) = \dfrac{z}{z^2+3z+2}$ in: ① ROC: $|z| > 3$, ② ROC: $|z| < \frac{1}{2}$ by using long division.

Solution:

① ROC: $|z| > 3$ الزمن موجب، لا يمكن تكون سالبة.

Poles: $z^2+3z+2 \Longrightarrow (z+2)(z+1)$

∴ Poles: $-2, -1$

$$z^{-1} - 3z^{-2} + 7z^{-3}$$

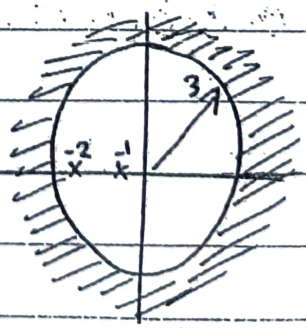
$$z^2+3z+2 \;\big|\; z$$

$$\mp z \mp 3 \mp 2z^{-1}$$

$$-3 - 2z^{-1}$$

$$\pm 3 \pm 9z^{-1} \pm 6z^{-2}$$

$$7z^{-1} + 6z^{-2}$$



* الزمن موجب، أس الـ z سالب

$z^{-1}, z^{-2}, z^{-3}, \ldots$

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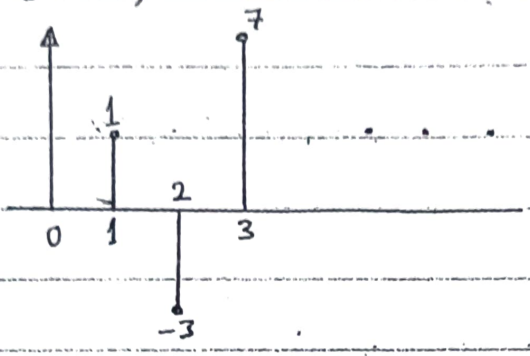
$X(z) = z^{-1} - 3z^{-2} + 7z^{-3} + \dots$

$= \delta(n-1) - 3\delta(n-2) + 7\delta(n-3)$



② ROC: $|z| < \frac{1}{2}$

* الإشارة سالب، الأسس موجبة

لأن $|z|$ أصغر

$$0.5z - 0.75z^2 + \frac{7}{8}z^3$$

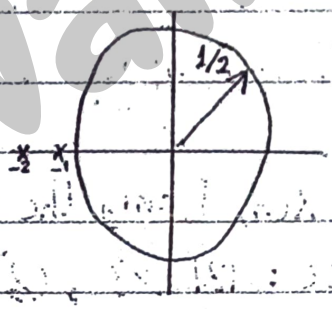
$$2 + 3z + z^2 \;\big|\; z$$

$$\mp z \mp 1.5z^2 \mp 0.5z^3$$

$$-1.5z^2 - 0.5z^3$$

$$\pm 1.5z^2 \pm 2.25z^3 \pm 0.75z^4$$

$$1.75z^3 + 0.75z^4$$



$X(z) = 0.5z - 0.75z^2 + 7/8\, z^3$

$= [\,7/8 \quad -0.75 \quad 0.5 \quad 0 \quad -\,]$

$X(n) = 0.5\delta(n+1) - 0.75\delta(n+2) + 7/8\,(n+3)$

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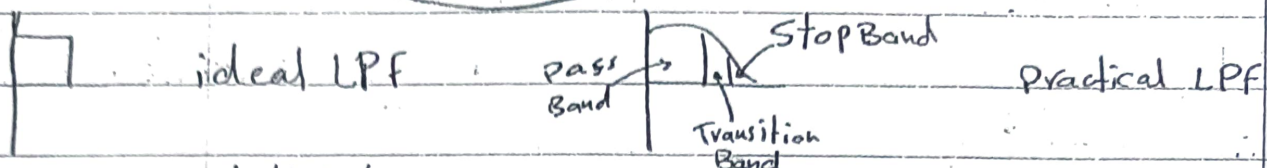
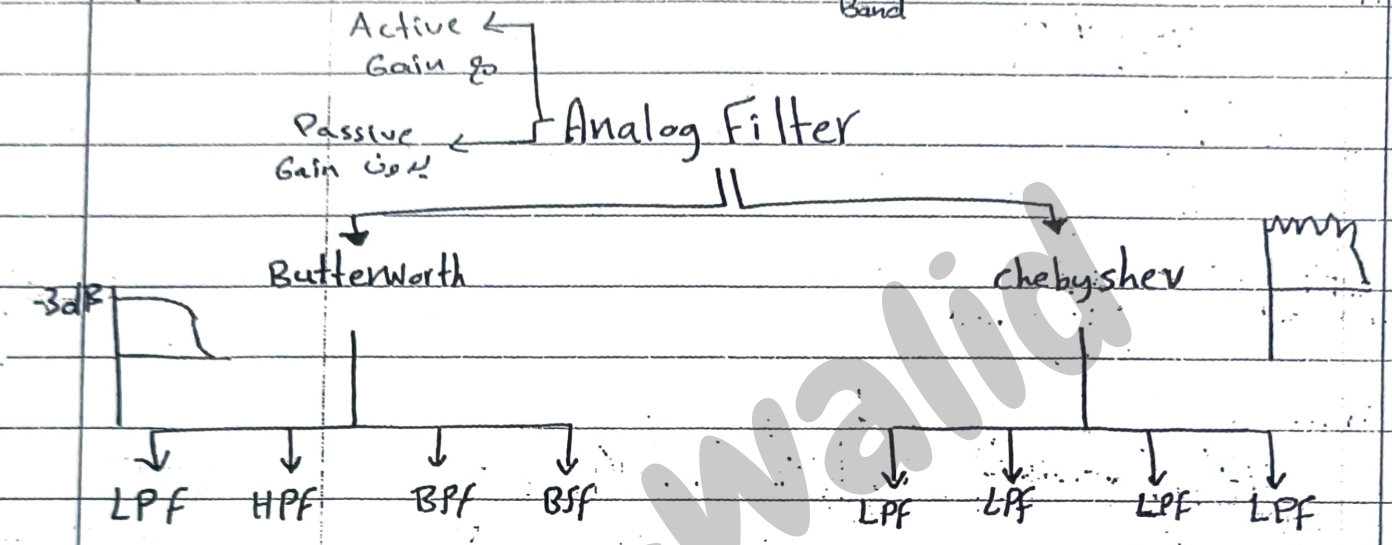
16/12/25

Chapter 3

Analog filter Design





Ex: Design analog Butterworth filter that has -2dB at Ω_1 20 rad/sec (cutoff freq), (at least) -10 dB at Ω_2 30 rad/sec (stopBand freq).

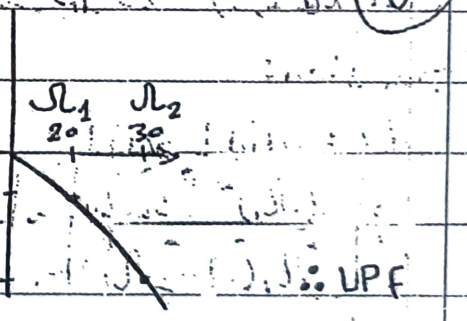
Solution:

$$n = \frac{\log\left[(10^{+K/10}-1)/(10^{-K/10}-1)\right]}{2\log\left(\frac{1}{\Omega_r}\right)}$$

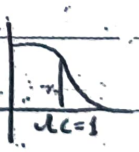
$\Omega_r > 1$

ratio of Ω_1 & Ω_2

$\Omega_r = 30/20 = 1.5$ (الأكبر على الصغير)



$$n = \frac{\log\left[(10^{+2/10}-1)/(10^{+10/10}-1)\right]}{2\log\left(\frac{1}{1.5}\right)} = 3.37 \approx 4$$ (تقريب للجدول)



$$H(s) = \frac{1}{(s^2+0.76s+1)(s^2+1.84s+1)} \quad B(s)$$

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$$\Omega_c = \frac{\Omega_1}{(10^{-K/10}-1)^{1/2n}} \quad \text{OR} \quad \Omega_c = \frac{\Omega_2}{(10^{-K2/10}-1)^{1/2n}}$$

Ω_1 ✓ $\qquad$ K2 & at least ✓'s *

$$\therefore \Omega_c = \frac{20}{(10^{0.2}-1)^{1/(2*4)}} = 21.38 \quad (\text{new } \Omega_c)$$

$$S \rightarrow \frac{S}{\Omega_c}$$

LPF normalized $\longrightarrow$ LPF with required cutoff Ω_c new

$$H(s) = \frac{1}{\left[\left(\frac{S}{21.38}\right)^2 + 0.76\left(\frac{S}{21.38}\right) + 1\right]\left[\left(\frac{S}{21.38}\right)^2 + 1.84\left(\frac{S}{21.38}\right) + 1\right]}$$

$S = j\Omega$ $\quad H(j\Omega)$ complex $\rightarrow |H(j\Omega)|$ pass

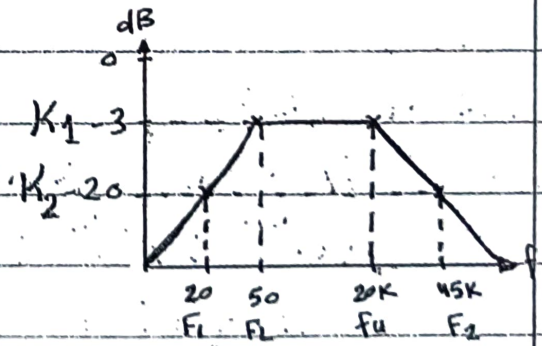
$|H(j\Omega)|$

3.41) Ex: design analog filter with -3dB at 50 Hz & 20 KHz (at least) -20 dB at 20 Hz & 45 KHz. monotonic. $\longrightarrow$ Butterworth.

Solution:

$$\Omega_r = \min[A, B]$$

$$A = \left|\frac{-(f_1)^2 + f_u f_L}{f_1(f_u - f_L)}\right| = \left|\frac{-(20)^2 + (50*20*10^3)}{20*(20*10^3 - 50)}\right| = 2.5$$



$$B = \left|\frac{-(\Omega_2)^2 + \Omega_L \Omega_u}{\Omega_2(\Omega_u - \Omega_L)}\right| = \left|\frac{-(f_2)^2 + f_L f_u}{f_2(f_u - f_L)}\right|$$

$$= \left|\frac{-(45*10^3) + (10^6)}{45*10^3(19950)}\right| = 2.25$$

$$\therefore \Omega_r = \min[2.5, 2.25] = 2.25$$

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17/1/3

$$n = \frac{\log_{10}\left[\left(10^{-K_1/10}-1\right)/\left(10^{-K_2/10}-1\right)\right]}{2\log\left(\frac{1}{\Omega_r}\right)} = \frac{\log_{10}\left[\left(10^{0.3}-1\right)/\left(10^{2}-1\right)\right]}{2\log\left(\frac{1}{2.25}\right)} = 2.8 \simeq 3$$

Butterworth Table

$$H(s) = \frac{1}{s^3+2s^2+2s+1}$$

$$s \longrightarrow \frac{s^2+\Omega_L\Omega_u}{s(\Omega_u-\Omega_L)} = \frac{s^2-4\pi^2*10^6}{s(2\pi*19950)} \qquad * \Omega_x = 2\pi f_x$$

$$W^\circ H(s) = \frac{1}{\left[\frac{s^2-4\pi^2*10^6}{s(2\pi*19950)}\right]^3+2\left[\frac{s^2-4\pi^2*10^6}{s(2\pi*19950)}\right]^2+2\left[\frac{s^2-4\pi^2*10^6}{s(2\pi*19950)}\right]+1}$$

BPF ↑

Ex: design analog filter that has:

① -3 dB at 20 Hz & 45 KHz

② -20 dB at 50 Hz & 20 KHz

③ Monotonic response.

↳ Butterworth

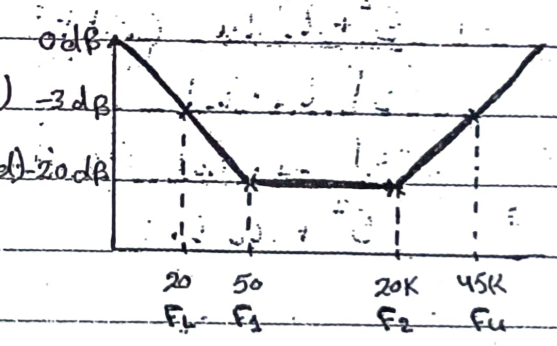
Solution:

$$A = \left|\frac{\Omega_1[\Omega_u-\Omega_L]}{-(\Omega_1)^2+\Omega_L\Omega_u}\right| = \left|\frac{f_1[f_u-f_L]}{-(f_1)^2+f_Lf_u}\right|$$

$$A = \left|\frac{50(45K-20)}{-(50)^2+(20*45K)}\right| = 2.5$$

$$B = \left|\frac{\Omega_2[\Omega_u-\Omega_L]}{-(\Omega_2)^2+\Omega_L\Omega_u}\right| = \left|\frac{f_2[f_u-f_L]}{-(f_2)^2+f_Lf_u}\right|$$

$$B = \left|\frac{20K(45K-20)}{-(20K)+45K*20}\right| = 2.25$$



BSF

$$\Omega_r = \min[A, B] = 2.25$$

$$n = \frac{\log\left[\left(10^{-K_1/10}-1\right)/\left(10^{-K_2/10}-1\right)\right]}{2\log\left(\frac{1}{\Omega_r}\right)} = \frac{\log\left[\left(10^{0.3}-1\right)/\left(10^{2}-1\right)\right]}{2\log\left(\frac{1}{2.25}\right)} = 2.8 \simeq 3$$

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$$H(s) = \frac{1}{s^3 + 2s^2 + 2s + 1}$$

$45K - 20 = 44980$

$45K * 20 = 900K$

$$s \longrightarrow \frac{s[2\pi * 44980]}{s^2[4\pi^2 * 900K]}$$

$$H(s) = \frac{1}{\left[\frac{s[2\pi*44980]}{s^2[2\pi^2*900K]}\right]^3 + 2\left[\frac{s[2\pi*44980]}{s^2[2\pi*900K]}\right]^2 + 2\left[\frac{s[2\pi*44980]}{s^2[2\pi*900K]}\right] + 1}$$

$$s \longrightarrow \frac{s}{\Omega_c} \quad \text{(LPF)}$$

$$s \longrightarrow \frac{\Omega_c}{s} \quad \text{(HPF)}$$

$$s \longrightarrow \frac{s^2 + \Omega_L \Omega_u}{s[\Omega_u - \Omega_L]} \quad \text{(BPF)}$$

$$s \longrightarrow \frac{s[\Omega_u - \Omega_L]}{s^2 + \Omega_L \Omega_u} \quad \text{(BSF)}$$

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17/1/8

Chebyshev filter

$$n = \frac{\log\left[g + \sqrt{g^2 - 1}\right]}{\log\left[\lambda_r + \sqrt{\lambda_r^2 - 1}\right]}$$

$$A = 10^{-K_2/20}$$

$$\epsilon = \left(10^{-K_1/10} - 1\right)^{1/2}$$

$$g = \sqrt{\frac{A^2 - 1}{\epsilon^2}}$$

Ex: Design a chebychev filter that has:

① acceptable Passband ripple of 2dB

② cutoff frequency 40 rad/sec

③ Stopband attenuation 20dB (or more) at 52 rad/sec.

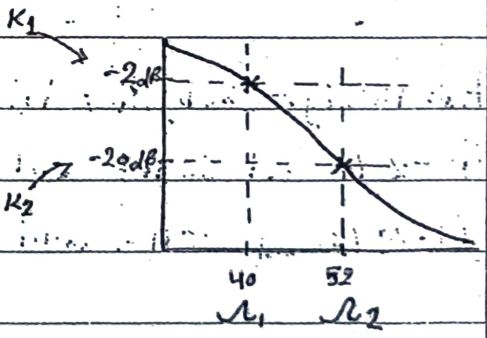
Solution:

$$\lambda_r = \frac{\lambda \text{ كبير}}{\lambda \text{ صغير}} = \frac{52}{40} = 1.3$$

$$A = 10^{+20/20} = 10$$

$$\epsilon^2 = \left(10^{+0.2} - 1\right) = 0.58$$

يمكن التحقق منها من الجدول مباشرة



LPF

$$g = \sqrt{\frac{10^2 - 1}{0.58}} = 13$$

$$n = \frac{\log\left[13 + \sqrt{13^2 - 1}\right]}{\log\left[1.3 + \sqrt{1.3^2 - 1}\right]} = 4.3 \sim 5$$

← تذهب الى جدول Chebyshev 20dB

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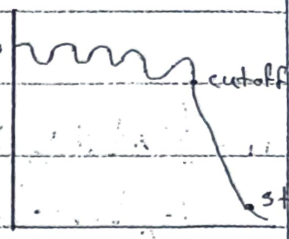
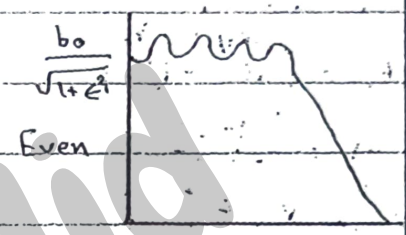
$$H(s) = \frac{K}{V(s)}, \quad K = \begin{cases} \frac{b_0}{\sqrt{1+\epsilon^2}} & n \text{ (even)} \\ b_0 & n \text{ (odd)} \end{cases}$$



$$V(s) = s^n + b_{n-1}s^{n-1} + \cdots + b_1 s + b_0$$

$$H(s) = \frac{0.08}{s^5 + 0.75s^4 + 1.49s^3 + 0.69s^2 + 0.45s + 0.08}$$



LPF (normalize) $\longrightarrow$ LPF (wanted)

$$s \longrightarrow \frac{s}{\Omega_c}$$

$\Omega_c \longrightarrow$ = Ω_1 (cutoff frequency)

$$s \longrightarrow \frac{s}{40}$$

$$H(s) = \frac{0.08}{\left(\frac{s}{40}\right)^5 + 0.7\left(\frac{s}{40}\right)^4 + 1.49\left(\frac{s}{40}\right)^3 + 0.69\left(\frac{s}{40}\right)^2 + 0.45\left(\frac{s}{40}\right) + 0.08}$$

Ex: Design Chebyshev filter that has:

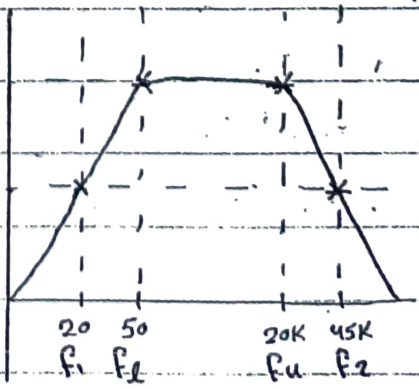
① -2dB at 50 Hz and 20KHz.

② -20dB at 20Hz and 45KHz.

Solution:

$$A = \left|\frac{-(f_1)^2 + f_L f_u}{f_1(f_u - f_L)}\right| = \left|\frac{-(20)^2 + (20K \times 50)}{20(20K - 50)}\right| = 2.5$$

$$B = \left|\frac{-(f_2)^2 + f_L f_u}{f_2(f_u - f_L)}\right| = \left|\frac{-(45K)^2 + (20K * 50)}{45K * (20K - 50)}\right| = 2.25$$



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$\therefore \Omega_s = 2.25$

$A = 10^{+20/20} = 10$

$\epsilon^2 = (10^{+0.2} - 1) = 0.58$

$$q = \sqrt{\frac{10^2 - 1}{0.58}} = 13$$

$$n = \frac{\log.\left[13 + \sqrt{13^2 - 1}\right]}{\log.\left[2.25 + \sqrt{2.25^2 - 1}\right]} = 2.24 \sim 3$$

$$H(s) = \frac{0.32}{s^3 + 0.73s^2 + 1.02s + 0.32}$$

$$s \longrightarrow \frac{s^2 + 4\pi^2 f_L f_u}{s[2\pi(f_u - f_L)]}$$

$$H(s) = \frac{0.32}{\left(\frac{s^2 + 4\pi^2 10^6}{2\pi * 19950}\right)^3 + 0.73\left(\frac{s^2 + 4\pi^2 10^6}{2\pi * 19950}\right)^2 + 1.02\left(\frac{s^2 + 4\pi^2 10^6}{2\pi * 19950}\right) + 0.32}$$

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Chapter 4
Digital filter design

1- Bilinear Method for IIR filter

Ex: design a digital LPF Butterworth that has:
-3dB at 0.5π rad (ω_1)
-15dB at 0.75π rad (ω_2)
use Bilinear Method.

وحدة قياس تردد الـ Digital. ✗ rad

وحدة قياس تردد الـ Analog. rad/sec

Solutions:

$$\Omega_1 = \frac{2}{T}\tan\left(\frac{\omega_1}{2}\right)$$

(ω_1: Digital freq., Ω_1: Analog freq.)

assume $T = 1$ لأنه غير موجودة بالسؤال

$$\Omega_1 = 2\tan\left(\frac{0.5\pi}{2}\right) = 2\tan(45) = 2 \text{ rad/sec}$$

$$\Omega_2 = 2\tan\left(\frac{0.75\pi}{2}\right) = 4.82 \text{ rad/sec}$$

$$n = \frac{\log\left[(10^{-K_1/10}-1)/(10^{-K_2/20}-1)\right]}{2\log\left(\frac{1}{\Omega_{1,2}}\right)}$$

$$\Omega_r = \frac{\Omega \text{ كبير}}{\Omega \text{ صغير}} = \frac{4.82}{2} = 2.41$$

$$n = \frac{\log\left[(10^{0.3}-1)/(10^{1.5}-1)\right]}{2\log\left(\frac{1}{2.41}\right)} = 1.94 \simeq 2$$

نذهب لجدول Butterworth

$$H(s) = \frac{1}{s^2 + \sqrt{2}\,s + 1}$$

normalized (LPF)

7/11/15

$S \rightarrow \frac{S}{\Omega_c}$

$\Omega_c = \frac{\Omega_1}{(10^{-K_1/10}-1)^{1/2n}} = \frac{2}{(10^{0.3}-1)^{1/2*2}} \approx 2$

$\therefore S \rightarrow \frac{S}{2}$

$H(s) = \frac{1}{(\frac{S}{2})^2 + \sqrt{2}(\frac{S}{2}) + 1}$

$S \rightarrow \frac{2}{T}\left(\frac{z-1}{z+1}\right)$, $T=1$

$H(z) = \frac{1}{(\frac{z-1}{z+1})^2 + \sqrt{2}(\frac{z-1}{z+1}) + 1}$

$H(z)|_{z=e^{j\omega}} = H(e^{j\omega})$ → $|H(e^{j\omega})|$, $|H(e^{j\omega})|$

Frequency Response

$H(z) = \frac{Y(z)}{X(z)}$ F.Z.T → $y(n)$, $x(n)$ difference equation

Ex: Design Butterworth digital LPF to be used in A/D-H(z)-D/A structure, which is equivalent to analog filter with $W = \Omega T$ (rad = (rad/sec)(sec))

-3dB at 500 Hz

-15dB at 750 Hz

T = 1/2000 sec

using Bilinear Transformation

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Solution:-

$w_1 = \Omega_1 T = 2\pi f_1 T$

$w_1 = 2\pi * 500 * 1/2000 = 0.5\pi$ rad

$w_2 = 2\pi f_2 T = 2\pi * 750 * 1/2000 = 0.75$ rad

$\Omega_1' = 2 \tan\left(\frac{w_1}{2}\right) = 2 \tan\left(\frac{0.5}{2}\right) = 2$ rad/sec

$\Omega_2' = 2 \tan\left(\frac{w_2}{2}\right) = 2 \tan\left(\frac{0.75}{2}\right) = 4.82$ rad/sec

$\Omega_r = \frac{\Omega_2'}{\Omega_1'} = \frac{4.82}{2} = 2.41$

$n = \frac{\log\left[\left(10^{\frac{-K_1}{10}} - 1\right) / \left(10^{\frac{-K_2}{10}} - 1\right)\right]}{2 \log\left(\frac{1}{\Omega_r}\right)}$

.

.

.

$H(z) = \frac{1}{\left(\frac{z-1}{z+1}\right)^2 + \sqrt{2}\left(\frac{z-1}{z+1}\right) + 1}$

Digital filter

Infinite Impulse Response (IIR) — Finite Impulse Response (FIR)

Bilinear — Impulse Invariant — Minimization of mean square (MSE)

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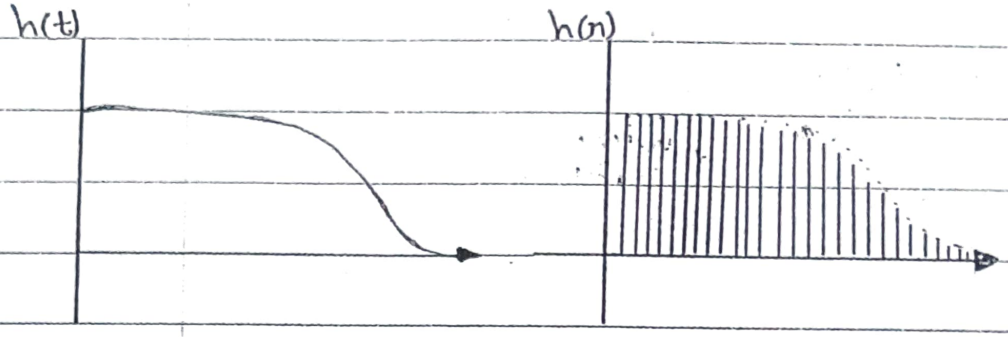
2- Impulse Invariant Method for IIR filter.



$$H_a(s) = \sum_{K=1}^{N} \frac{A_K}{S+\alpha_K} = \frac{A_1}{S+\alpha_1} + \frac{A_2}{S+\alpha_2} + \frac{A_3}{S+\alpha_3} + \dots + \frac{A_N}{S+\alpha_N}$$

using Partial Fraction

تحويل الى Z

$$H(Z) = \sum_{K=1}^{N} \frac{A_K Z}{Z - e^{-\alpha_K T}} = \frac{A_1 Z}{Z - e^{-\alpha_1 T}} + \frac{A_2 Z}{Z - e^{-\alpha_2 T}} + \dots + \frac{A_N Z}{Z - e^{-\alpha_N T}}$$

Ex: Design a Digital filter using Impulse Invariant method, that has:

$$H(s) = \frac{1}{s^2+3s+2}, \quad T = 0.1 \text{ sec}$$

Solution :-

$$H(s) = \frac{A}{(s+2)} + \frac{B}{(s+1)}$$

$$A = \lim_{s \to -2} \frac{1}{(s+1)} = \boxed{-1} \qquad B = \lim_{s \to -1} \frac{1}{(s+2)} = \boxed{1}$$

$$H(s) = \frac{1}{(s+1)} - \frac{1}{(s+2)}$$

$$H(z) = \frac{1 * z}{z - e^{-1*0.1}} - \frac{1*z}{z - e^{-2*0.1}} = \frac{z}{z-0.9} - \frac{z}{z-0.8}$$

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3- Minimization of mean Square (MSE) for IIR filter

$$E = \sum_{i=1}^{M} \left[|H(e^{jwi})| - |H_d(e^{jwi})| \right]^2$$

Error — H_d: desired (المطلوب)

FIR filter design using window method:-

Window	W_t	K_2(dB)
Rectangle	$4\pi/N$	-21
Bartlett	$8\pi/N$	-25
Hanning	$8\pi/N$	-44
Hamming	$8\pi/N$	-53
Blackman	$12\pi/N$	-74

W_t = Transition width = $W_2 - W_1$

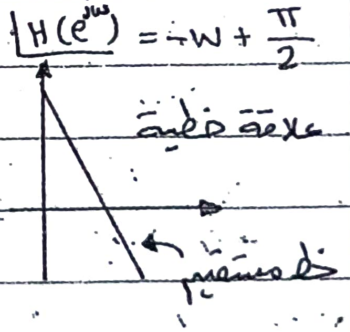
K_2(dB) = Stopband attenuation

Ex: Design LPF (digital) to be used in A/D - H(z) - D/A structure that has: ($W = \Omega T$; digital ← W → analog)

① -3dB at 30π (Ω_1) rad/sec

② -50dB at 45π (Ω_2) rad/sec

③ linear phase with $T = 1/100$ sec → يعني طريقة FIR, window



Solution:-

$\therefore K_2 = -50$ dB $\therefore$ Window is Hamming

$W_1 = \Omega_1 T = 30\pi * 1/100 = 0.3\pi$ rad

$W_2 = \Omega_2 T = 45\pi * 1/100 = 0.45\pi$ rad

$W_t = 0.45\pi - 0.3\pi = 0.15\pi$ rad

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$Wt = 8\pi/N \Rightarrow N = 8\pi/Wt = 8\pi/0.15\pi = 53.3 \simeq \boxed{55}$

Linear Phase عدد العينات N يجب أن يكون عدد صحيح فردي ليصبح

$d = \frac{N-1}{2} = \frac{55-1}{2} = 27$

Linear Phase ليكون filter ال shift بـ

$h(n) = \frac{\sin[w_c(n-d)]}{\pi(n-d)}$ (IIR)

$h(n) = h(n) \cdot w(n)$ (FIR) (IIR) (window)

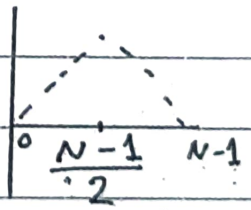
$w_c = w_1 = 0.3\pi$ Hamming window

$\therefore h(n) = \frac{\sin[0.3\pi(n-27)]}{\pi(n-27)} \cdot \left[0.54 - 0.46 \cos \frac{2\pi n}{(N-1)}\right]$ (FIR) (55)

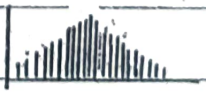
① Rectangle $w(n) = 1$ $0 \leq n \leq N-1$

② Bartlett $w(n)$

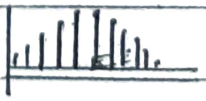
$\begin{cases} \frac{2n}{N-1} & 0 \leq n \leq \frac{N-1}{2} \\ 2 - \frac{2n}{N-1} & \frac{N-1}{2} \leq n \leq N-1 \end{cases}$



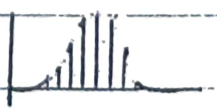
③ Hanning $w(n)$ $\left[0.5 - 0.5 \cos \frac{2\pi n}{N-1}\right]$



④ Hamming $w(n)$ $\left[0.54 - 0.46 \cos \frac{2\pi n}{N-1}\right]$



⑤ Blackman $w(n)$ $\left[0.42 - 0.5 \cos \frac{2\pi n}{N-1} + 0.08 \cos \frac{4\pi n}{N-1}\right]$



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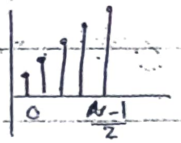
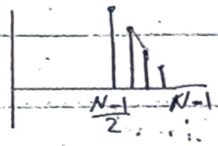
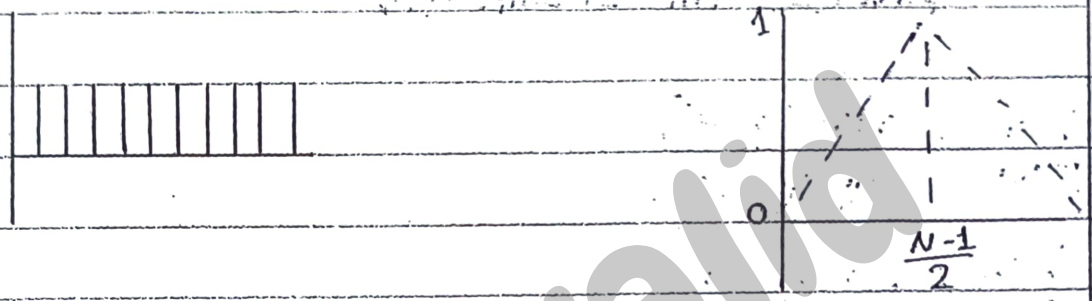
2017/1/2

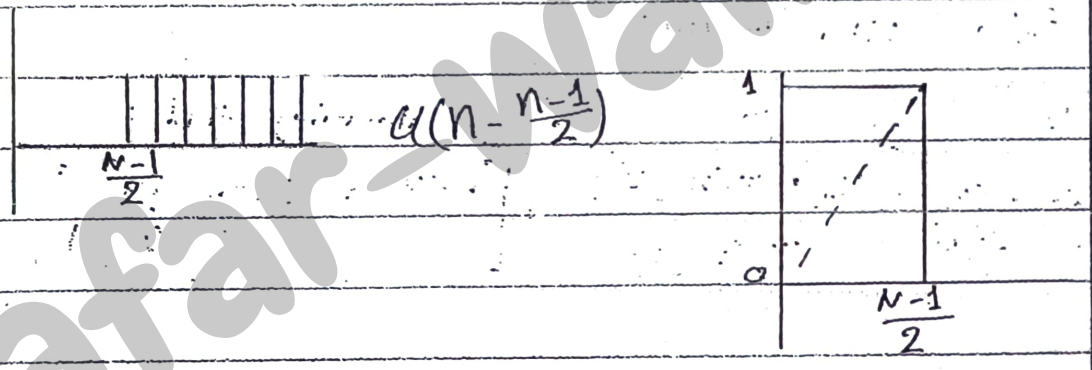
Bartlett Window:

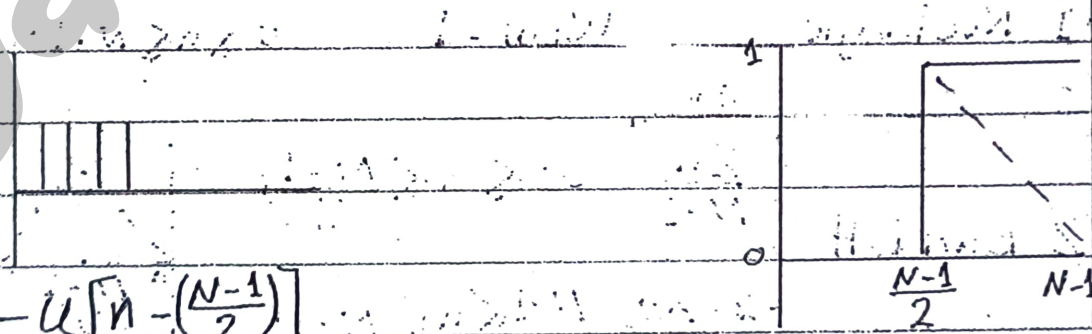
$$W(n) = \begin{cases} \frac{2n}{N-1} & 0 \leq N \leq \frac{N-1}{2} \\ 2 - \frac{2n}{N-1} & \frac{N-1}{2} \leq n \leq (N-1) \end{cases}$$



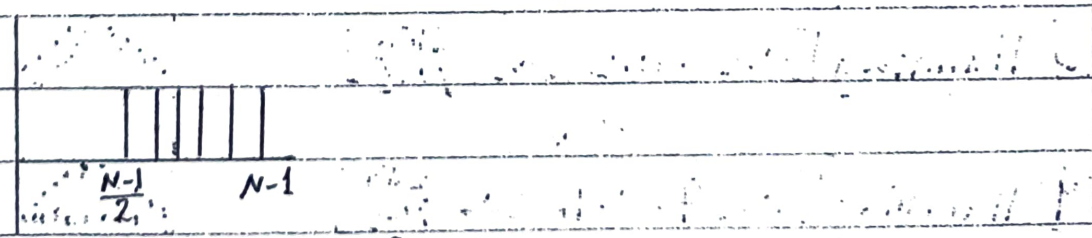








$$u(n) - u\left[n - \left(\frac{N-1}{2}\right)\right]$$



$$u\left(n - \left(\frac{N-1}{2}\right)\right) = u\{n - (N-1)\}$$

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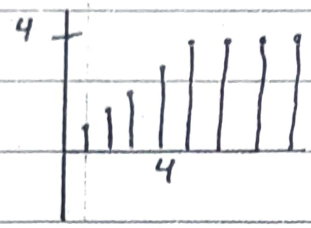
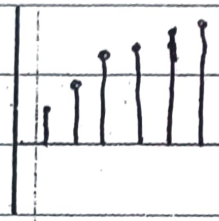
$$\therefore W(n) = \frac{2n}{N-1}\left\{u(n) - u\left[n-\left(\frac{N-1}{2}\right)\right]\right\} + \left\{\left(2-\frac{2n}{N-1}\right)\cdot\left[u\left(n-\left(\frac{N-1}{2}\right)\right) - u(n-(N-1)\right.\right.$$

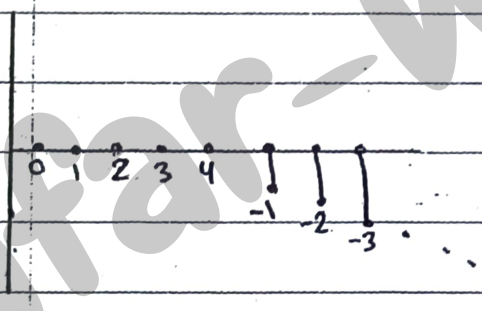


$u(n) = 1 \quad 0 \leq n$

$r(n) = n\ u(n)$

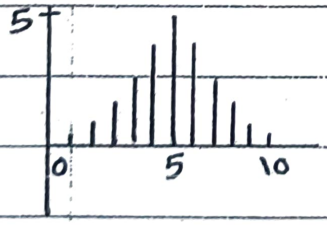


$r(n) = n\ u(n)$



$= -(n-4)\ u(n-4)$

$X(n)_{\text{Total}} = n\,u(n) - (n-4)\,u(n-4)$



$X(t) = t\,u(t) - 2(t-5)u(t-5) + (t-10)\,u(t-10)$

at $t=1$

$X(1) = 1*u(1) - 2(1-5)\,u(1-5) + (1-10)\,u(1-10)$

at $t=6$

$X(6) = 6\,u(6) - 2(6-5)\,u(6-5) + (6-10)\,u(6-10)$

$= 6-2 = 4$

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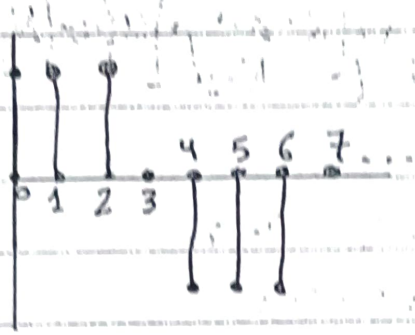

$X(n) = u(n) - u(n-3) - u(n-4) + u(n-7)$

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Chapter 5
Realization of Digital Filter

17/3/15

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1} a_k z^{-k}}$$

$$Y(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}} \quad (M \leq N)$$

$a_0 = 1$ (always)

$$Y(z) + a_1 Y(z) z^{-1} + \dots + a_N Y(z) z^{-N} = b_0 X(z) + b_1 X(z) z^{-1} + \dots + b_M X(z) z^{-M}$$

Taking inverse Z-Transform (Z-Domain → Time Domain)

Difference Equation

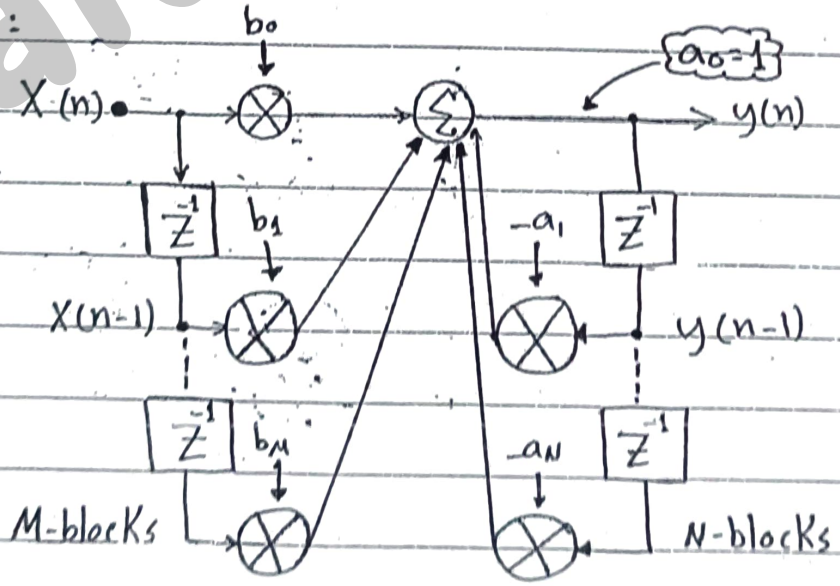
$$y(n) + a_1 y(n-1) + \dots + a_N y(n-N) = b_0 X(n) + b_1 X(n-1) + \dots + b_M X(n-M)$$

① Direct Form I

Ex: Find The Realization of Digital Filter of IIR using Direct Form I

$$y(n) = b_0 X(n) + b_1 X(n-1) + \dots + b_M X(n-M) - a_1 y(n-1) + \dots + a_N y(n-N)$$

Solution:



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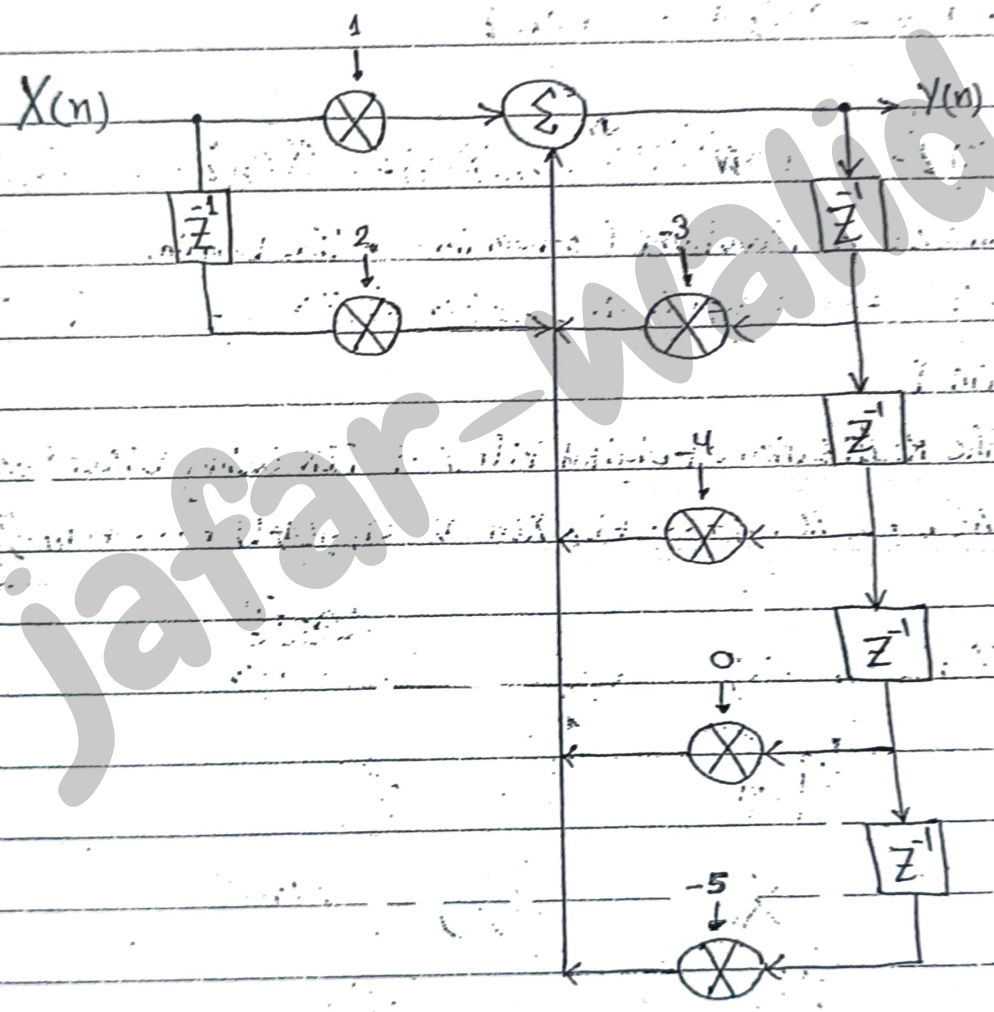
Ex: Find Direct Form I Realization For:

$$H(z) = \frac{b_0 + b_1 z^{-1}}{1 + 3z^{-1} + 4z^{-2} + 5z^{-4}} = \frac{1 + 2z^{-1}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + a_4 z^{-4}}$$

Solution:



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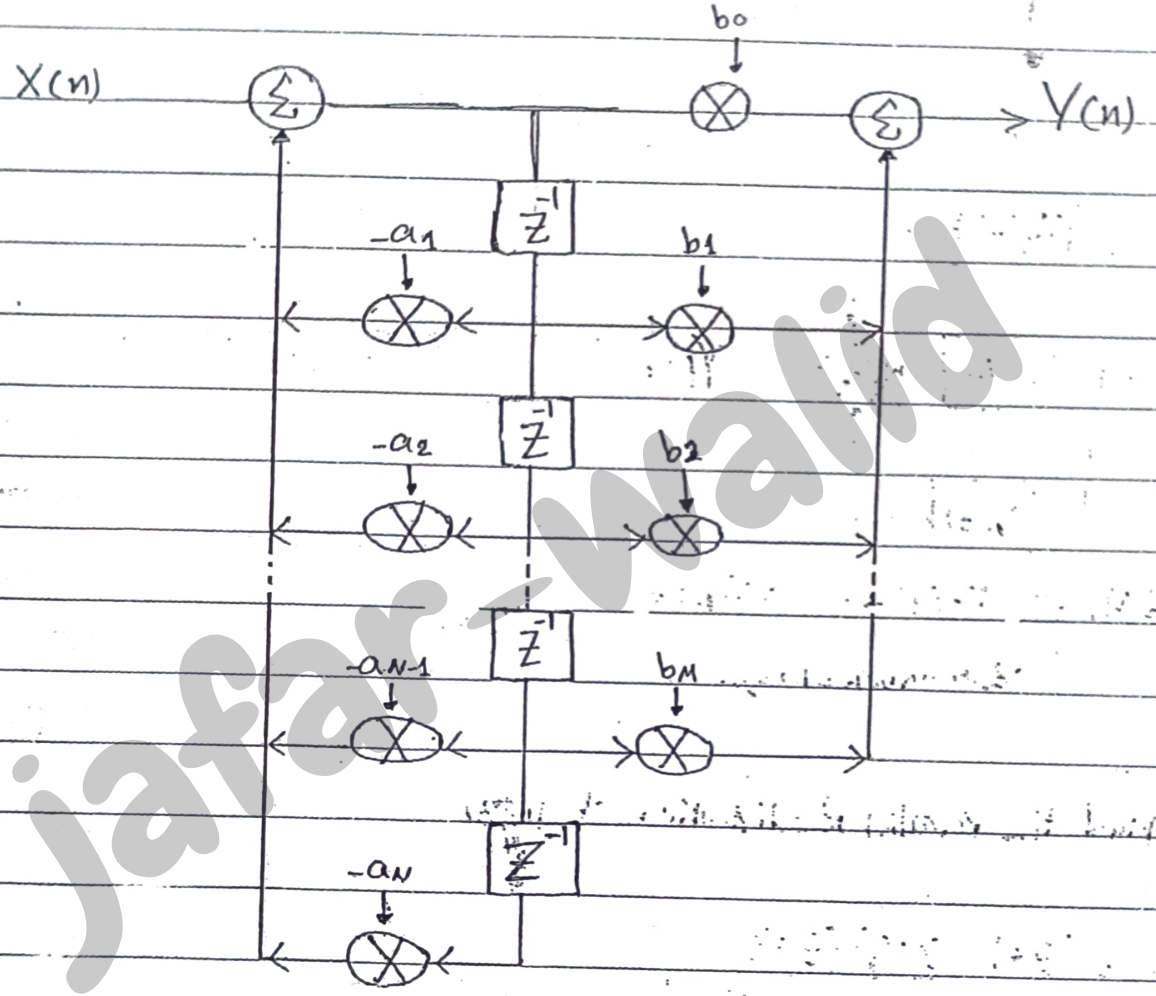
② Direct Form II

* في هذه الصيغة بلوكات الـ z^{-1} تكون مشتركة بين X(n) و Y(n)، وبذلك يكون عدد هذه البلوكات أقل.



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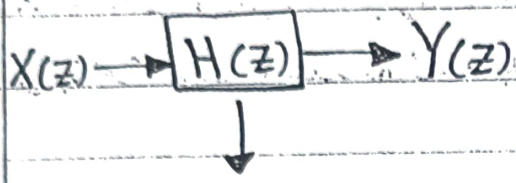
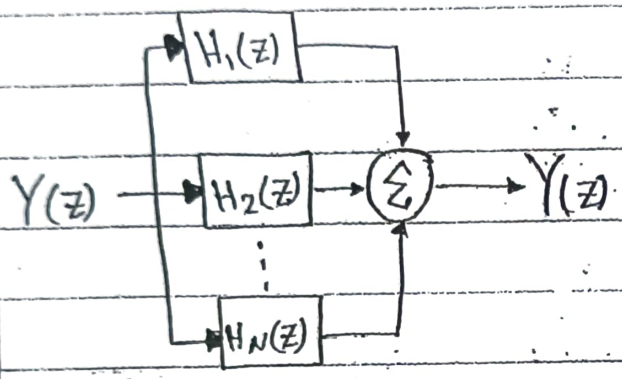
3) Parallel Realization





$H(z) = H_1(z) + H_2(z) + \cdots + H_N(z)$

Zero ORDER always.

Ex: Find the Parallel Realization of H(z)

$$H(z) = \frac{(z+1)(z+2)(z+3)}{(z+4)(z+5)(z+6)}$$

Solution:

Partial fraction:

$$H(z) = \frac{A}{(z+4)} + \frac{B}{(z+5)} + \frac{C}{(z+6)}$$

$$A = \lim_{z \to -4} \frac{(-3)*(-2)*(-1)}{1*2} = -3$$

$$B = \lim_{z \to -5} \frac{(-4)*(-3)*(-1)}{-1*1} = 24$$

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$$C = \lim_{z \to -6} \frac{(-5)*(-4)*(-3)}{(-2)*(-1)} = -30$$

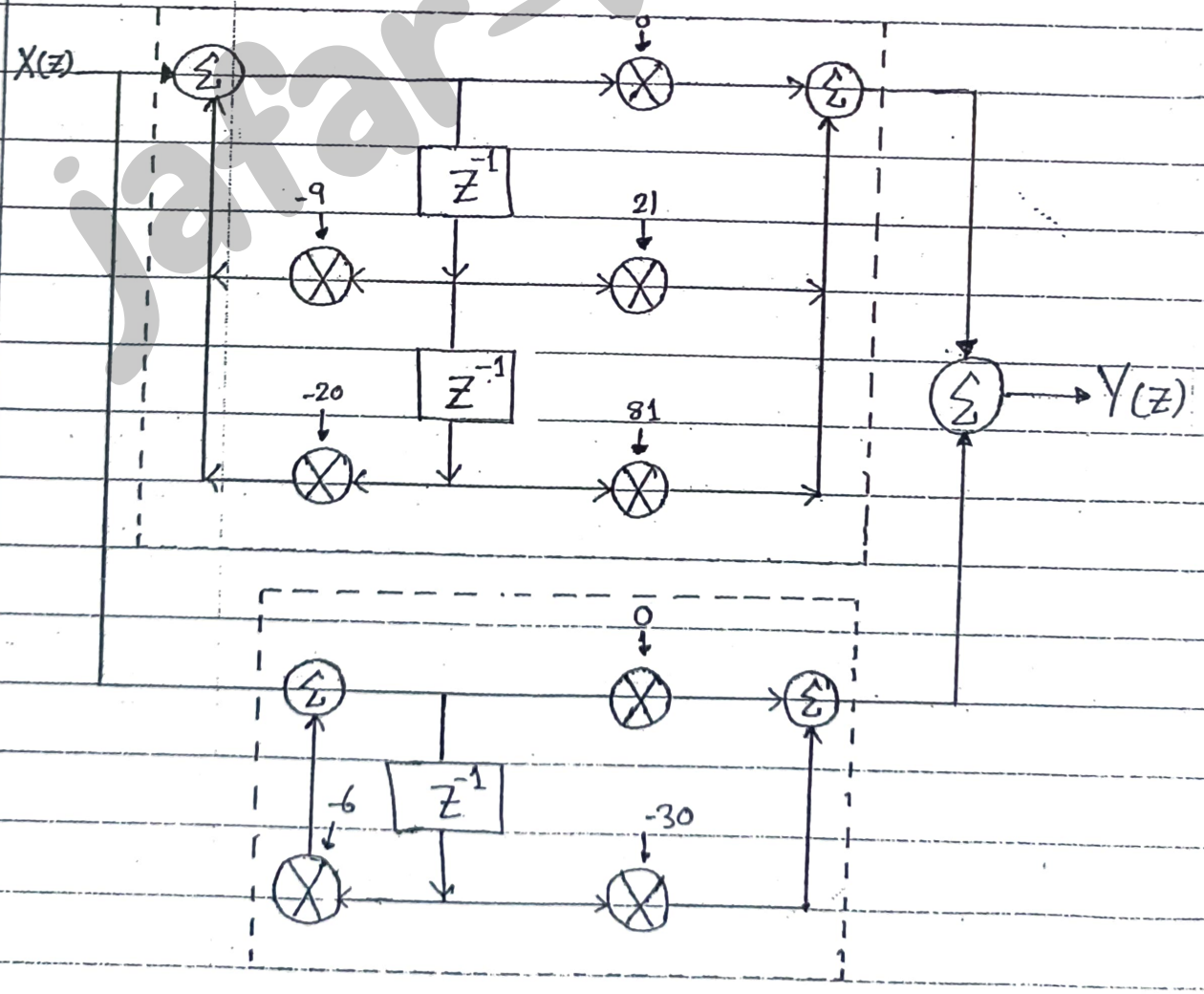
$$H(z) = \frac{-3}{z+4} + \frac{24}{z+5} - \frac{30}{z+6} = \frac{-3z-15+24z+96}{z^2+9z+20} - \frac{30}{z+6}$$

$$= \frac{21z+81}{z^2+9z+20} - \frac{30}{z+6}$$

$H_1(z) * \frac{z^{-2}}{z^{-2}}$ $\qquad$ $H_2(z) * \frac{z^{-1}}{z^{-1}}$

$$H_1(z) = \frac{21z^{-1}+81z^{-2}}{1+9z^{-1}+20z^{-2}} \qquad H_2(z) = \frac{-30z^{-1}}{1+6z^{-1}}$$

using Canonical Form (Direct Form II)



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Chapter 6

Discrete Fourier Transform (DFT)

$$X(K) = \sum_{n=0}^{N-1} X(n)\, e^{-jw_o nK} \qquad 0 \le K \le N-1$$

$$X(n) = \frac{1}{N} \sum_{K=0}^{N-1} X(K)\, e^{jw_o nK} \qquad 0 \le K \le N-1$$

$X(K) = DFT\{X(n)\}$

$X(n) = IDFT\{X(K)\}$

$$w_o = \frac{2\pi}{N}$$

Ex: find $\{X(K)\}$ if $X(n) = [1\ 2\ 3\ 4]$ → DFT

Solution:

$N = 4$

$$w_o = \frac{2\pi}{N} = 2\pi/4 = \frac{\pi}{2}$$

at $K = 0$ $\quad X(0) = \sum_{n=0}^{3} X(n)\, e^{0}$

$X(0) = X(0) + X(1) + X(2) + X(3)$

$= 1 + 2 + 3 + 4 = 10$

at $K = 1$ $\quad X(1) = X(0)\, e^{-j\pi/2 * 0} + X(1)\, e^{-j\pi/2 * 1} + X(2)\, e^{-j\pi/2 * 2} + X(3)\, e^{-j\pi/2 * 3}$

$= 1 + (-2j) + (-3) + (4j)$

$= -2 + 2j$

$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$

$e^{-j\pi/2} = -j$ $\quad (-90)$

$e^{j\theta}$ $\quad \angle\theta$

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NEXUS GUY

at K=2 $X(2) = \sum_{n=0}^{3} X(n)\, e^{-j\frac{2\pi}{4}\times 2\times n}$

$= \sum_{n=0}^{3} X(n)\, e^{-j\pi n}$

$= X(0)\, e^{j0} + X(1)\, e^{-j\pi} + X(2)\, e^{-j2\pi} + X(3)\, e^{-j3\pi}$

$= 1 + (-2) + (+3) + (-4) = -2$

at K=3 $X(3) = \sum_{n=0}^{3} X(n)\, e^{-j\frac{2\pi}{4}\times 3\times n}$

$\sum_{n=0}^{3} X(n)\, e^{j\frac{3\pi}{2} n}$

$X(3) = -2 - j2$

Matrix Method:

$$\begin{bmatrix} X(K) \\ (4\times 1) \end{bmatrix} = \begin{bmatrix} W \\ (4\times 4) \end{bmatrix} \times \begin{bmatrix} X(n) \\ (4\times 1) \end{bmatrix}$$

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \times \begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix}$$

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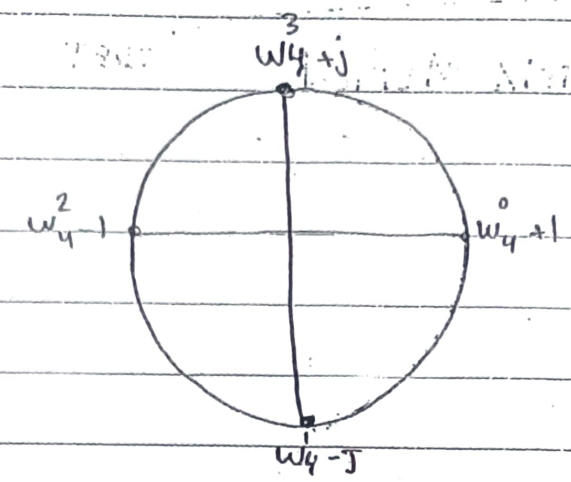
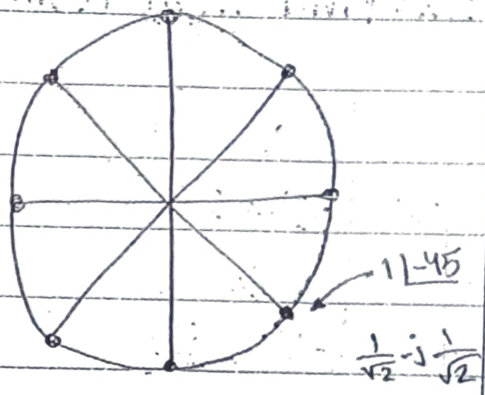


$N = 4$



$N = 8$

$W_4^0 = W_4^4 = W_4^8$

$W_4^9 = W_4^5 = W_4^1$

Ex: find $X(K)$ using Matrix Method (DFT) $x(n) = [1\ 2\ 3\ 4]$

Solution:

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 \\ -2+j2 \\ -2 \\ -2-j2 \end{bmatrix}$$

Inverse Matrix Method:

$$\frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} X(K) \\ X(K) \\ X(K) \\ X(K) \end{bmatrix}$$

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Ex: find $x(n)$ using Matrix Method — IDFT

$$X(K) = \begin{bmatrix} 10 \\ -2+2j \\ -2 \\ -2-2j \end{bmatrix}$$

Solution:

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 10 \\ -2+j2 \\ -2 \\ -2-j2 \end{bmatrix}$$

$$X(0) = \frac{1}{4}[10-2+j2-2-2-2j] = 1$$

$$X(1) = \frac{1}{4}[10-2j-2+2+2j-2] = 2$$

$$X(2) = \frac{1}{4}[10+2-j2-2+2+j2] = 3$$

$$X(4) = \frac{1}{4}[10+2j+2+2-2j+2] = 4$$

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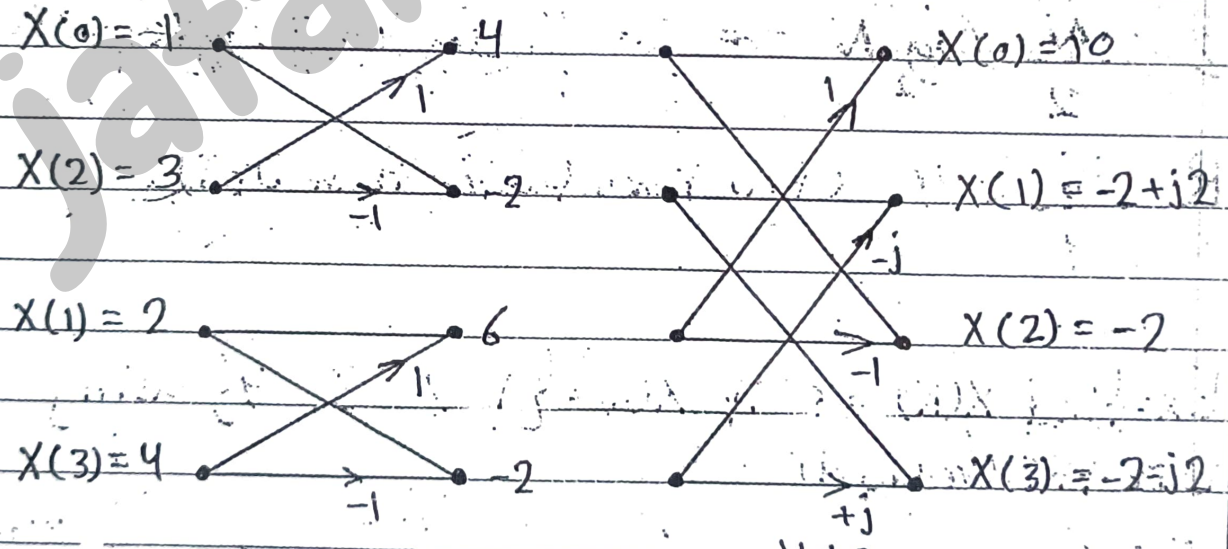
Fast Fourier Transform (FFT)

- DIT (Time)
  - ① DIT-FFT
  - ② DIT-FFT (reduced) (Cooley-Tukey)
- DIF (Frequency)
  - ③ DIF-FFT
  - ④ DIF-FFT (reduced) (Cooley-Tukey)

Ex: find X(K) (DFT) using DIT-FFT, if $X(n) = [1\ 2\ 3\ 4]$

Solution

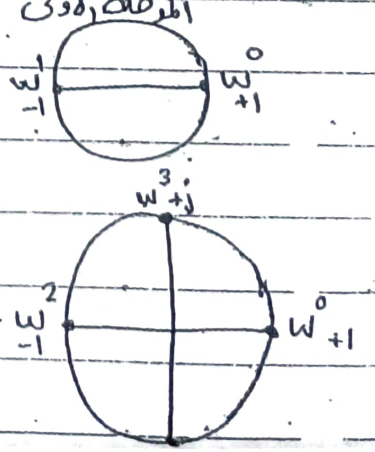
$N = 2^{\nu} \Rightarrow 4 = 2^{\nu} \rightarrow \therefore \nu = 2$ (عدد المراحل; عدد العينات)



Dec n	Bin	Rev-Bin	Dec
0	00	00	0
1	01	10	2
2	10	01	1
3	11	11	3

العنوان الثنائية

المرحلة الأولى



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Ex: find ξ and R for DIT-FFT and DIT-FFT reduced, if N=4.

Solution:

matrix $\longrightarrow \xi = N^2 = 4^2 = 16$

for DIT-FFT:

$\xi = N \log_2 N \Longrightarrow \xi = 4 \log_2 4 = 4 \log_2 (2)^2 = 8 \log_2 2 = 8$

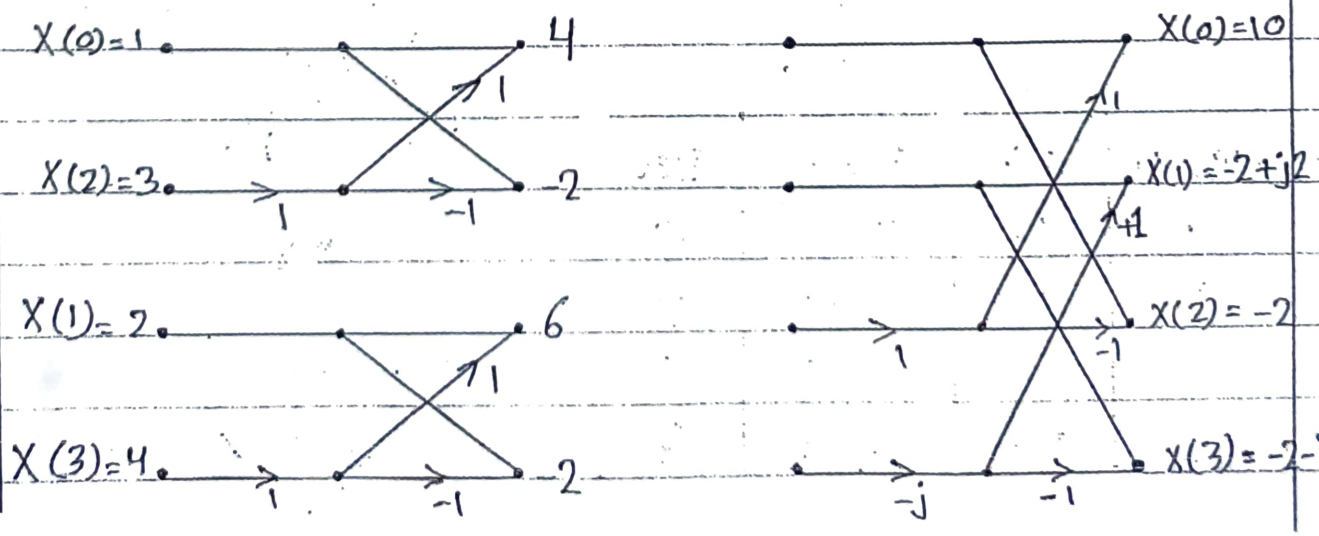
$R = \frac{16}{8} = 2$ (twice faster than DFT)

for DIT-FFT reduced:

$\xi = \frac{N}{2} \log_2 N \Longrightarrow \xi = \frac{4}{2} \log_2 4 = 2 \log_2 (2)^2 = 4$

$R = \frac{16}{4} = 4$ (four times faster than DFT)

Ex: find X(K) = ? if $X(n) = \{1\ 2\ 3\ 4\}$ using DIT-FFT (reduced)



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Inverse FFT

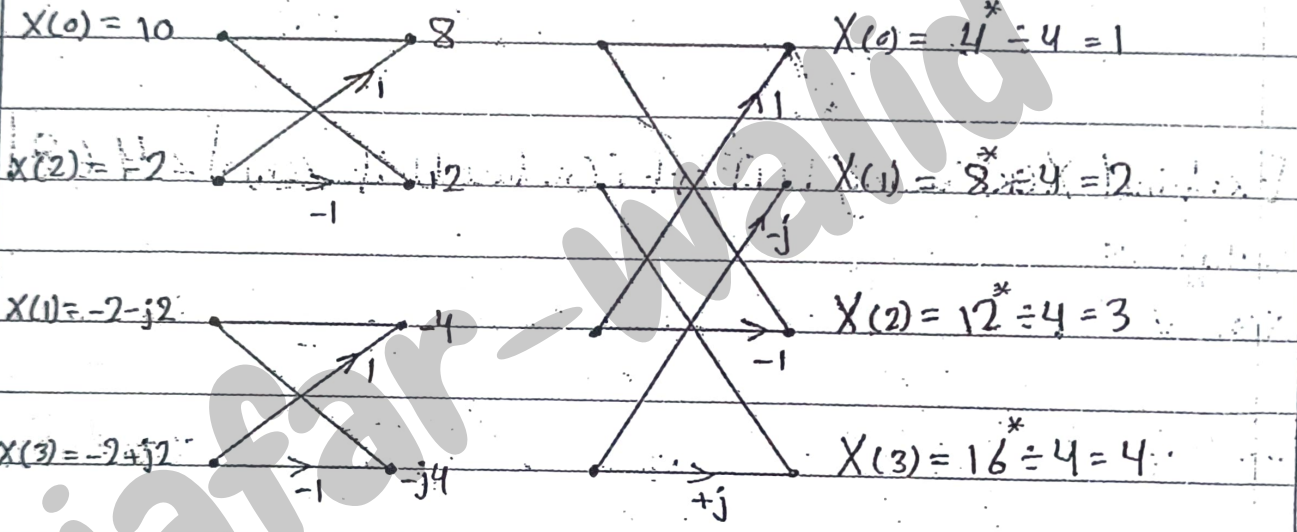
$$x(n) = \frac{1}{N}\left[FFT\{X^*(k)\}\right]^*$$

(Time عينة المجزأة)

Ex: find x(n) using DIT-FFT, if X(k) = [10 -2+j2 -2 -2-j2]

Solution:

N = 4

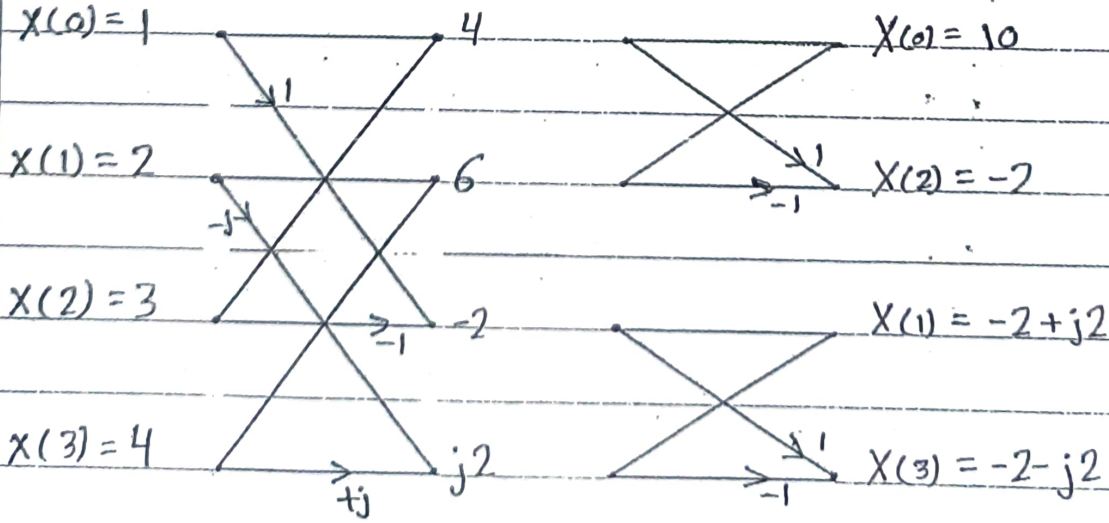


Ex: find X(k) using DIF-FFT, if x(n) = [1 2 3 4]

Solution:

(Freq. عينة المجزأة)

N = 4



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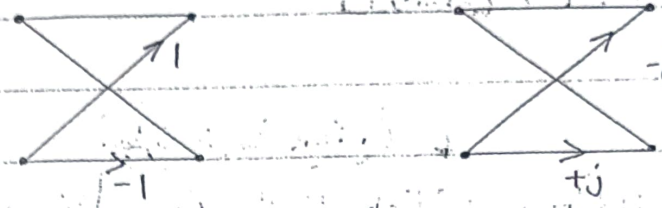
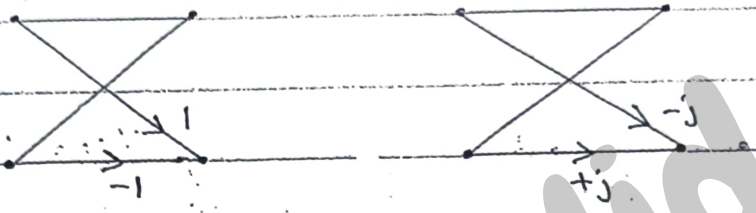
* Conjugate of a real number is the Same number

DIT

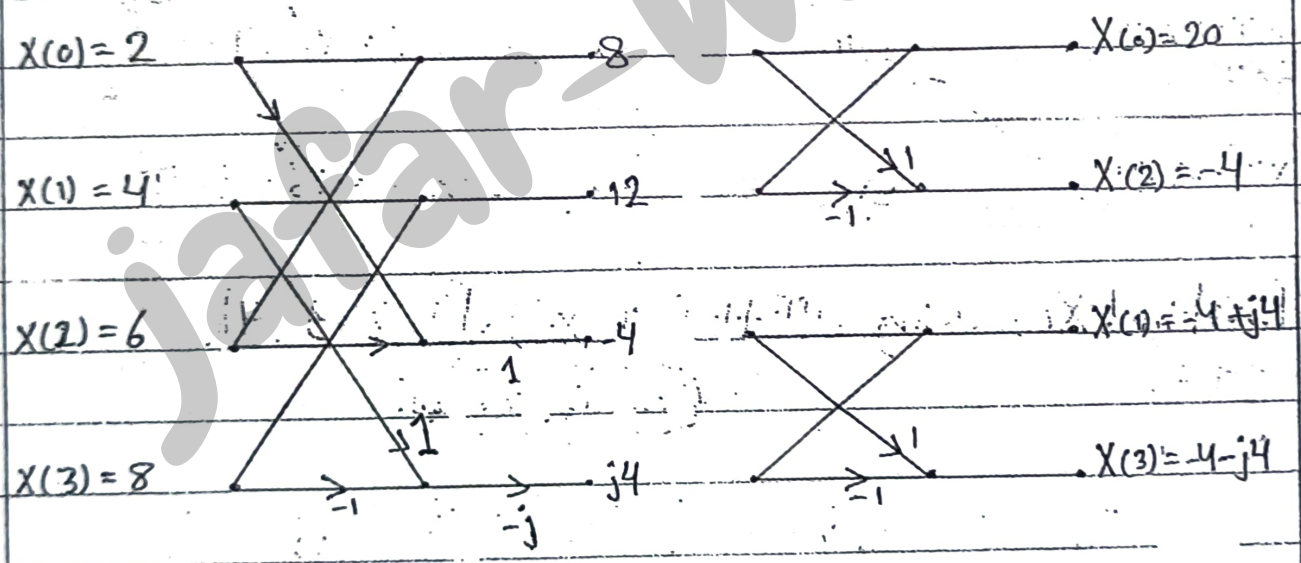


DIF



Ex: find X(K) using DIF-FFT (reduced) if x(n) = [2 4 6 8]

Solution:



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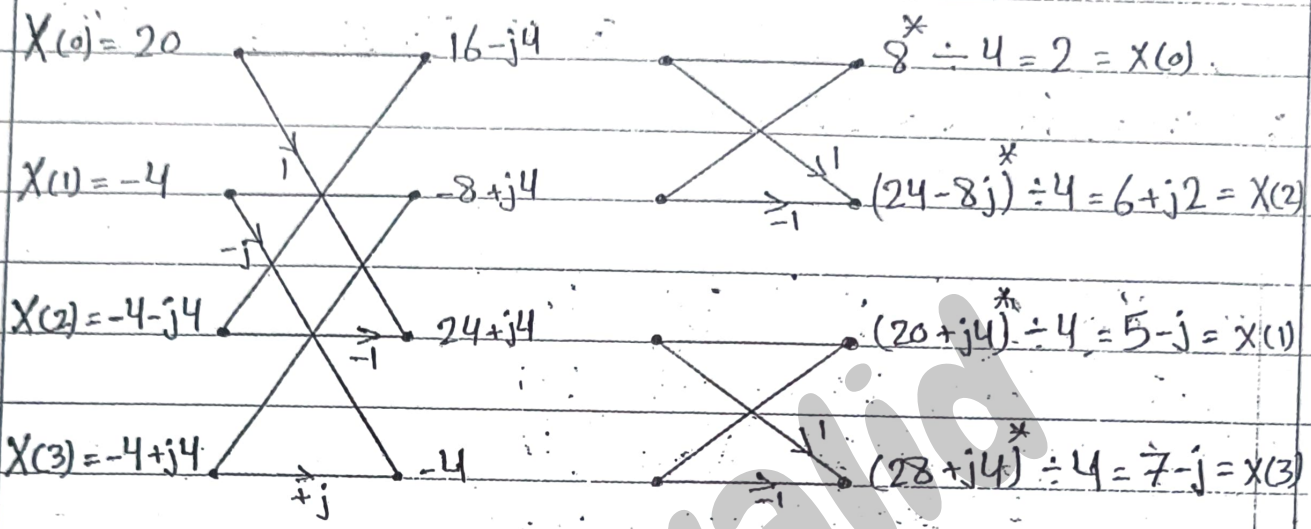
Ex: find x(n) using DIF-FFT, if $X(k)=[20\ -4\ -4+j4\ -4-j4]$

Solution:



Ex: find for $X_1(n)=[1\ 2\ 3\ 4]$ $X_2(n)=[5\ 6\ 7\ 8]$:-

① Circular Convolution $y(n) = X_1(n) \circledast X_2(n)$

② Linear Convolution $y(n) = X_1(n) * X_2(n)$

Solution:

① let n=0 $X_1(n)=[1\ 2\ 3\ 4]$

نعكس الإشارة $X_2(n)=[5\ 8\ 7\ 6]$

W_1 بنت $5\ 16\ 21\ 24 = 66 = y(0)$

let n=1 $X_1(n)=[1\ 2\ 3\ 4]$

$X_2(n)=[6\ 5\ 8\ 7]$

$6\ 10\ 24\ 28 = 68 = y(1)$

let n=2 $X_1(n)=[1\ 2\ 3\ 4]$

$X_2(n)=[7\ 6\ 5\ 8]$

$7\ 12\ 15\ 32 = 66 = y(2)$

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let n = 3 $X_1(n) = [1\ 2\ 3\ 4]$

$X_2(n) = [8\ 7\ 6\ 5]$

$8\ 14\ 18\ 20 = 60 = y(3)$

$\therefore y(n) = [66\ 68\ 66\ 60]$

2)

x_2 \ x_1	1	2	3	4
5	(5)	10	15	20
6	6	12	18	(24)
7	7	14	21	28
8	8	(16)	(24)	(32)

$y(n) = [5\ 16\ 34\ 60\ 61\ 52\ 32]$

N = 4

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الفصل الأول — قوانين DSP

$$H(e^{jw}) = \sum_{K=-\infty}^{\infty} h(K)\, e^{-jwK} \qquad K=n$$

$H(e^{jw})$: Frequency Responce — $h(K)$: Impulse Responce

$$y_{ss}(n) = A * |H(e^{jwo})| \overset{\sin}{\cos}[W_o n + \varphi + \underline{H(e^{jwo})}]$$

$$X(n) = A * \overset{\sin}{\cos}[W_o n + \varphi]$$

* اذا واجهنا متسلسلة سنفصل $H(e^{jw})$ ونستخدم $e^{\pm j\theta} = \cos\theta \pm j\sin\theta$.

* اذا لم نواجه متسلسلة سنأخذ $|H(e^{jw})|$, $\underline{H(e^{jw})}$ مباشرة من $H(e^{jw})$.

$$\underbrace{1(n)}_{\text{pulse espance}} = \frac{1}{2\pi}\int_{-\pi}^{\pi} \underbrace{H(e^{jw})}_{\text{Frequency Responce}}\, e^{jwn}\, dw$$

$$w = \Omega T \quad , \quad t = nT$$

rad / s — rad

$$\left[\frac{e^{jx} + e^{-jx}}{2}\right] = \cos x \quad , \quad \left[\frac{e^{jx} - e^{-jx}}{2j}\right] = \sin x$$

$$\frac{\sin(x)}{x} = \text{Sinc}(x)$$

$$y(n) = a\,x(n) \Longrightarrow h(n) = a\,\delta(n)$$

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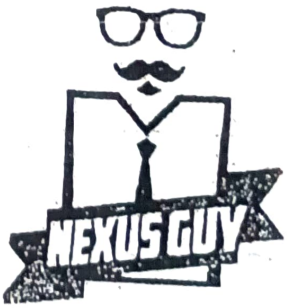

Chapter 3

Butter Worth Filter :-

$$n = \frac{\log_{10}\left[\left(10^{-K_1/10} - 1\right) / \left(10^{-K_2/10} - 1\right)\right]}{2\log_{10}\left(\frac{1}{\Omega_r}\right)}$$

ow Pass – High Pass Filters :

$$\Omega_r = \frac{\Omega\ (\text{الكبير})}{\Omega\ (\text{الصغير})}$$

Butter Worth
chebyshev

Band Pass - Band Stop Filters:

$$\Omega_r = \min[A, B]$$

Butterworth
chebyshev

and Pass Filter:

$$= \left|\frac{-(f_1)^2 + f_2 f_u}{f_1(f_u - f_L)}\right| \qquad B = \left|\frac{-(f_2)^2 + f_2 f_u}{f_2(f_u - f_L)}\right|$$

Butterworth
chebyshev

and Stop Filter:

$$= \left|\frac{f_1(f_u - f_L)}{-(f_1)^2 + f_L f_u}\right| \qquad B = \left|\frac{f_2(f_u - f_L)}{-(f_2)^2 + f_u f_L}\right|$$

Butter Worth
chebyshev

w Pass - High Passfilters:

$$c = \frac{\Omega}{\left(10^{-K/10} - 1\right)^{1/2n}}$$

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Chebyshev and Butterworth:

$$S \longrightarrow \frac{S}{\Omega_c} \quad \text{(LPF)}$$

$$S \longrightarrow \frac{\Omega_c}{S} \quad \text{(HPF)}$$

$$S \longrightarrow \frac{S^2+\Omega_L\Omega_u}{S[\Omega_u-\Omega_L]} \quad \text{(BPF)}$$

$$S \longrightarrow \frac{S[\Omega_u-\Omega_L]}{S^2+\Omega_L\Omega_u} \quad \text{(BSF)}$$

Bilinear and Butterworth:

$$\Omega_1 = \frac{2}{T}\tan\left(\frac{\omega_1}{2}\right), \quad \Omega_2 = \frac{2}{T}\tan\left(\frac{\omega_2}{2}\right)$$

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Chebyshev Filter:

$$n = \frac{\log\left[g + \sqrt{g^2 - 1}\right]}{\log\left[\Omega_r + \sqrt{\Omega_r^2 - 1}\right]}$$

$$\epsilon = \sqrt{10^{-K_1/10} - 1}$$

$$g = \sqrt{\frac{A^2 - 1}{\epsilon^2}}$$

Chapter 4

FIR:

$$h(n)_{IIR} = \frac{\sin[w_c(n-d)]}{\pi(n-d)} \qquad d = \frac{N-1}{2}$$

$$h(n)_{FIR} = h(n)_{IIR} \cdot w(n)_{window}$$

$$* \Omega = \frac{2}{T} \tan\left(\frac{w}{2}\right)$$

T → Default = 1

$$\Delta t = w_2 - w_1$$

window	w(n)	W(t)	K_2(dB)
Rectangle	1	$4\pi/N$	-21
Bartlett	$2n/(N-1)$ $\quad 2 - 2n/(N-1)$	$8\pi/N$	-25
Hanning	$\left[0.5 - 0.5\cos\frac{2\pi n}{N-1}\right]$	$8\pi/N$	-44
Hamming	$\left[0.54 - 0.46\cos\frac{2\pi n}{N-1}\right]$	$8\pi/N$	-53
Blackman	$\left[0.42 - 0.5\cos\frac{2\pi n}{N-1} + 0.08\cos\frac{4\pi n}{N-1}\right]$	$12\pi/N$	-74

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$$X(K) = \sum_{n=0}^{N-1} X(n)\, e^{-jw_o nK}$$

$$X(K) = DFT(X(n))$$

$$X(n) = \frac{1}{N} \sum_{K=0}^{N-1} X(K)\, e^{+jw_o nK}$$

$$X(n) = IDFT(X(K))$$

$$* \; w_o = \frac{2\pi}{N}$$

Matrix Method

$$\underset{(4*1)}{X(K)} = \underset{(4*4)}{W} \times \underset{(4*1)}{X(n)}$$

$$\underset{(4*1)}{X(n)} = \frac{1}{4} \underset{(4*4)}{W} \times \underset{(4*1)}{X(K)}$$

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where $\oint_{C_1}$ is a complex contour integral and C_2 is a closed contour in the intersection of the ROCs of $X(v)$ and $Y(z/v)$. The region of convergence for the resulting transform is $R_{x-}R_{y-} < |z| < R_{x+}R_{y+}$.

7. *Initial Value Theorem*

If $x(n)$ is a causal sequence with $\mathcal{Z}$ transform $X(z)$, then

$$x(0) = \lim_{z\to\infty} X(z) \tag{2.8}$$

8. *Final Value Theorem*

If the $\mathcal{Z}[x(n)] = X(z)$ and the poles of $X(z)$ are all inside the unit circle, then the value of $x(n)$ as $n \to \infty$ is given by

$$\lim_{n\to\infty} x(n) = \lim_{z\to 1}\left[\left(\frac{z-1}{z}\right)X(z)\right] \tag{2.9}$$

The properties just presented allow the calculation of $\mathcal{Z}$ transforms of sequences formed by various operations on sequences whose transforms are known. For these properties to be useful, a catalog of transforms of fundamental or building sequences must be established. In the following discussion a minimal table will be determined for special positive and negative time sequences, while a more complete table for positive time sequences is given in Appendix C.

2.1.3. Transforms of Some Useful Sequences

In Section 2.1 the $\mathcal{Z}$ transform of a positive time exponential sequence was found. Using this result and the properties given in Section 2.1.2, the $\mathcal{Z}$ transforms of certain useful sequences, shown in Table 2.1, are easily established.

The first entry in the table is found by applying the definition [Eq. (2.1)] directly to the unit sample sequence. As the only nonzero value is at $n = 0$, the summation contains the single term 1, which converges everywhere. A unit sample sequence translated by an amount k to the right results in a negative power of z and thus we cannot have convergence at $z = 0$.

Entry 3 of Table 2.1 was obtained in Example 2.1. Also, since the unit step is just an exponential $a^n u(n)$ with $a = 1$, the $\mathcal{Z}$ transform of $u(n)$ can be written as $z/(z-1)$ with region of convergence $|z| > 1$, thus establishing entry 2 of Table 2.1. To prove the results of entry 4 requires breaking up the sinusoidal sequences into sums of complex exponential sequences, applying the linearity property given in Section 2.1.2 to the known transforms of the exponential sequences, and simplifying the complex expressions.

TABLE 2.1 TABLE OF THE z TRANSFORM OF A FEW USEFUL SEQUENCES

	Sequence	z Transform	Region of convergence
1. Unit sample	$\delta(n)$	1	All z
	$\delta(n-k)\quad k>0$	z^{-k}	$\lvert z\rvert > 0$
	$\qquad\quad k<0$	z^{-k}	$\lvert z\rvert < \infty$
2. Unit step	$u(n)$	$z/(z-1)$	$\lvert z\rvert > 1$
	$-u(-n-1)$	$z/(z-1)$	$\lvert z\rvert < 1$
3. Exponential	$a^n u(n)$	$z/(z-a)$	$\lvert z\rvert > \lvert a\rvert$
	$-b^n u(-n-1)$	$z/(z-b)$	$\lvert z\rvert < \lvert b\rvert$
4. Sinusoidal	$\sin \omega_0 n\, u(n)$	$\dfrac{z\sin\omega_0}{z^2 - 2z\cos\omega_0 + 1}$	$\lvert z\rvert > 1$
	$\cos \omega_0 n\, u(n)$	$\dfrac{z^2 - z\cos\omega_0}{z^2 - 2z\cos\omega_0 + 1}$	$\lvert z\rvert > 1$
5. Unit ramp	$nu(n)$	$\dfrac{z}{(z-1)^2}$	$\lvert z\rvert > 1$

The $\mathcal{Z}$ transform of the unit ramp can be obtained directly from the definition and series evaluation using summation formulae from Appendix B or by using the $\mathcal{Z}$ transform of the unit step sequence and property 4. Since $nu(n)$ is n times $u(n)$, property 4 gives the transform of $nu(n)$ as

$$\mathcal{Z}[nu(n)] = -z\frac{d}{dz}\mathcal{Z}[u(n)] = -z\frac{d}{dz}\left(\frac{z}{z-1}\right) = \frac{z}{(z-1)^2}$$

with the same region of convergence as that for the transform of the unit step sequence, i.e., $|z| > 1$.

The transforms of more complex analytical representations that can be decomposed into cascade operations can be easily found by repeated application of the properties in Section 2.1. The transforms shown in Table 2.1 are used as the basis as shown in the following example.

EXAMPLE 2.4

Find the $\mathcal{Z}$ transform of the following $x(n)$ by using the properties of the $\mathcal{Z}$ transforms and Table 2.1:

$$x(n) = (n-2)a^{(n-2)}\cos[\omega_0(n-2)]\,u(n-2)$$

Solution. Applying the translation property gives

$$\mathcal{Z}[x(n)] = z^{-2}\mathcal{Z}[na^n \cos \omega_0 n\, u(n)] \qquad \text{no change in ROC}$$

Applying the multiplication by a ramp property yields

$$\mathcal{Z}[x(n)] = z^{-2}\left\{-z\frac{d}{dz}\mathcal{Z}[a^n \cos \omega_0 n\, u(n)]\right\} \qquad \text{no change in ROC}$$

Using the multiplication by an exponential property results in

$$\mathcal{Z}[x(n)] = -z^{-1}\frac{d}{dz}\{\mathcal{Z}[\cos \omega_0 n\, u(n)]|_{z=z/a}\}$$

where the ROC is

$$|a|R_{x-} < |z| < |a|R_{x+}$$

The R_{x-} and R_{x+} above signify the region of convergence for the transform of $\cos \omega_0 n\, u(n)$, which is $|z| > 1$. Using Table 2.1 to obtain the Z transform of $\cos \omega_0 n\, u(n)$ yields

$$\mathcal{Z}[x(n)] = -z^{-1}\frac{d}{dz}\left(\frac{z^2 - z\cos \omega_0}{z^2 - 2z\cos \omega_0 + 1}\bigg|_{z=z/a}\right) \qquad \text{for } |z| > |a|$$

To arrive at the final answer requires replacing z by z/a, taking the derivative, and simplifying.

2.1.4 Regions of Convergence for Special Sequences

The region of convergence for the Z transform is very important. We have already seen that the Z transforms of $a^n u(n)$ and $-a^n u(-n-1)$ both give $z/(z-a)$, the only distinguishing characteristic being the region of convergence. The purpose of this section is to establish some general results for convergence of transforms of sums of right- and left-hand sequences.

Definition. A right-hand sequence $x(n)$ is one for which $x(n) = 0$ for all $n < n_0$, where n_0 is positive or negative but finite. If n_0 is greater than or equal to zero, the resulting sequence is causal or a positive time sequence.

Definition. A left-hand sequence $x(n)$ is one for which $x(n) = 0$ for all $n \geq n_0$, where n_0 is positive or negative but finite. If $n_0 \leq 0$, the resulting sequence is negative time or anticausal sequence.

Suppose $x(n)$ is a causal sequence that can be written as a sum of complex exponentials. This takes in a wide class of functions including sinusoids, exponentials, and products thereof. Let

$$x(n) = \sum_{i=1}^{N} (a_i)^n u(n)$$

Taking the Z transform of $x(n)$ gives

$$\mathcal{Z}[x(n)] = X(z) = \sum_{i=1}^{N} \frac{z}{z - a_i}$$

The region of convergence $\mathcal{R}$ is the intersection of the regions of convergence for each exponential as follows:

$$\mathcal{R} = \bigcap_{i=1}^{N} \mathcal{R}_i, \qquad \text{where } \mathcal{R}_i = \{z : |z| > |a_i|\}$$

Therefore, $\mathcal{R} = \{z : |z| > \text{largest of } |a_i|\}$ as shown in Fig. 2.4.

Since the region of convergence for a translated exponential remains the same as that for the original exponential Eq. (2.3), all right-hand sequences that are sums of translated exponentials have regions of convergence similar to that expressed above.

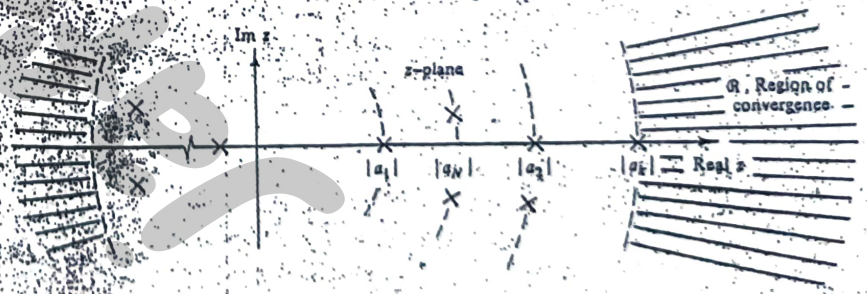


Figure 2.4 Region of convergence for the Z transform of the sum of causal complex exponential sequences.

terworth, Chebyshev, and elliptic low-pass filters are presented along with the procedures and transformations necessary to transform them into other low-pass, high-pass, bandpass, and bandstop filters.

3.1 BUTTERWORTH FILTERS

A linear time invariant analog filter can be characterized by its system function $H(s)$ or its corresponding frequency response $H(j\Omega)$. The Butterworth filter of order n is described by the magnitude squared of its frequency response given below:

$$|H_n(j\Omega)|^2 = 1/[1 + (\Omega/\Omega_c)^{2n}] \tag{3.1}$$

In Fig. 3.3 the magnitude squared frequency response of the Butterworth filter is shown for several different values of n.

The following properties are easily determined:

1. $|H_n(j\Omega)|^2|_{\Omega=0} = 1$ for all n.
2. $|H_n(j\Omega)|^2|_{\Omega=\Omega_c} = \frac{1}{2}$ for all finite n.
   This implies that
   $|H_n(j\Omega)|_{\Omega=\Omega_c} = 0.707$
   and $20 \log |H_n(j\Omega)|_{\Omega=\Omega_c} = -3.0103$.
3. $|H_n(j\Omega)|^2$ is a monotonically decreasing function of Ω.
4. As n gets larger, $|H_n(j\Omega)|^2$ approaches an ideal low-pass frequency response.
5. $|H_n(j\Omega)|^2$ is called maximally flat at the origin since all order derivatives exist and are zero.

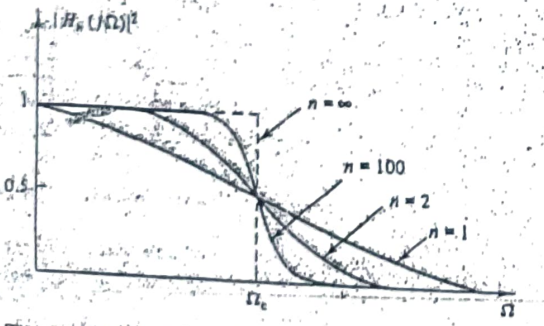


Figure 3.3 The magnitude squared frequency response for a Butterworth filter.

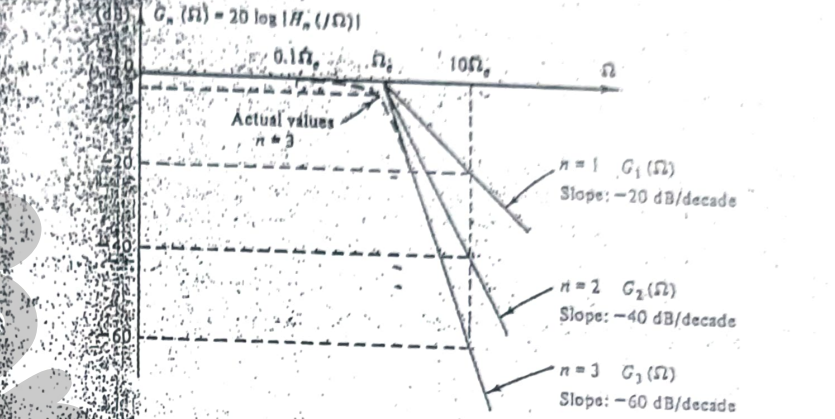


Figure 3.4 Filter gain plot for analog Butterworth filters of various orders n.

It is convenient in many cases to look at the frequency response in decibels, that is, plot $20 \log |H(j\Omega)|$ versus Ω. Figure 3.4 is a straight line approximation of the frequency response in decibels for the Butterworth filter. The straight lines are determined for $\Omega \ll \Omega_c$ and $\Omega \gg \Omega_c$ by approximating $20 \log |H_n(j\Omega)|$. We have

$$G_n(\Omega) = 20 \log |H_n(j\Omega)| = 10 \log |H_n(j\Omega)|^2$$
$$= 10 \log |1/[1 + (\Omega/\Omega_c)^{2n}]| = -10 \log [1 + (\Omega/\Omega_c)^{2n}] \tag{3.2}$$

Since the $(\Omega/\Omega_c)^{2n}$ is approximately zero for $\Omega \ll \Omega_c$, we have

$$G_n(\Omega) \approx -10 \log |1 + 0| = 0$$

For $\Omega \gg \Omega_c$, the 1 in the summation becomes insignificant compared to $(\Omega/\Omega_c)^{2n}$, so

$$G_n(\Omega) \approx -10 \log |(\Omega/\Omega_c)^{2n}| = -20n \log |\Omega/\Omega_c| \tag{3.3}$$

Therefore, the approximate gain for $\Omega \ll \Omega_c$ is 0 dB while for $\Omega \gg \Omega_c$ the slope of $G_n(\Omega)$ is $-20n$ dB/decade. For $\Omega = \Omega_c$ the actual value of $G_n(\Omega)$ is approximately -3 dB, as seen earlier, and the actual values for Ω around Ω_c can be calculated using (3.2). These values are shown by dotted lines in Fig. 3.4 for $n = 3$.

In most of the work to follow, the normalized low-pass Butterworth filter

will be considered, which is, the case for $\Omega_c = 1$ rad/sec. It will be shown, once the normalized low-pass transfer function has been determined, that the transfer function for any other Butterworth low-pass, high-pass, bandpass or bandstop filter can be obtained by applying a transformation to the normalized low-pass filter specified by $H_n(s)$.

Starting with the magnitude squared frequency response we would like to find the system function $H(s)$ that gives the Butterworth magnitude squared response. For an analog system we remember that the frequency response is obtained by letting $s = j\Omega$ in the transfer function $H(s)$ for the given system. Therefore, if Ω is replaced by s/j, the system function is determined. Setting $\Omega_c = 1$ in Eq. (3.1) gives $|H_n(j\Omega)|^2$ for the normalized filter as follows:

$$|H_n(j\Omega)|^2 = H_n(j\Omega)H_n(-j\Omega) = 1/(1 + \Omega^{2n})$$

$$H_n(s) \cdot H_n(-s) = \frac{1}{1 + (s/j)^{2n}} \tag{3.4}$$

The poles of $H_n(s) \cdot H_n(-s)$ are given by the roots of the denominator, i.e.,

$$1 + (s/j)^{2n} = 0$$

or

$$s^{2n} = -1(j)^{2n} = (-1)^{n+1} \tag{3.5}$$

The roots of the above equation can be identified for the cases when n is odd and even. For n odd, the poles of $H_n(s)H_n(-s)$ become the $2n$th roots of 1, while for n even, the poles are the $2n$th roots of -1. That is,

for n odd: $s_k = 1\angle k\pi/n, \quad k = 0, 1, 2, \ldots, 2n-1$

for n even: $s_k = 1\angle \pi/2n + k\pi/n, \quad k = 0, 1, 2, \ldots, 2n-1$

These poles for n odd and even are illustrated in Fig. 3.5. For n odd we have a pole of $H_n(s) \cdot H(-s)$ at $s = 1$, and then poles equally spaced on the unit circle π/n in angle, while for n even the first pole is at $1\angle \pi/2n$ and the remaining poles equally spaced around the unit circle by π/n.

If we wish the filter $H_n(s)$ to be a stable and causal filter, the poles of $H_n(s)$ are selected to be those in the left half plane and $H_n(s)$ can be written in the following form:

$$H_n(s) = \frac{1}{\prod_{\text{LHP poles}} (s - s_k)} = \frac{1}{B_n(s)} \tag{3.6}$$

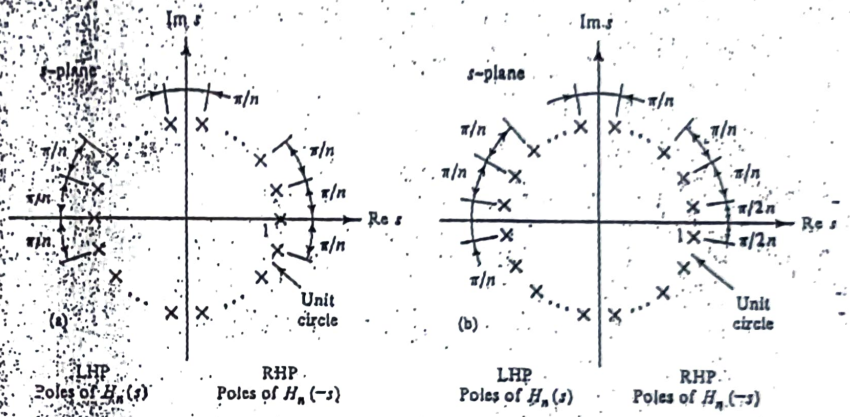


Figure 3.5 Poles of $H_n(s)H(-s)$ for a low-pass Butterworth filter with a 1-rad cutoff frequency. (a) n odd. (b) n even.

where s_k are all the left half plane poles of $H_n(s)\,H_n(-s)$. The denominator, $B_n(s)$, can be shown to be a Butterworth polynomial of order n. Table 3.1a gives the first five Butterworth polynomials in a real factored form.

The following examples find the first and second order Butterworth polynomials and corresponding filter transfer functions $H_1(s)$ and $H_2(s)$.

EXAMPLE 3.1

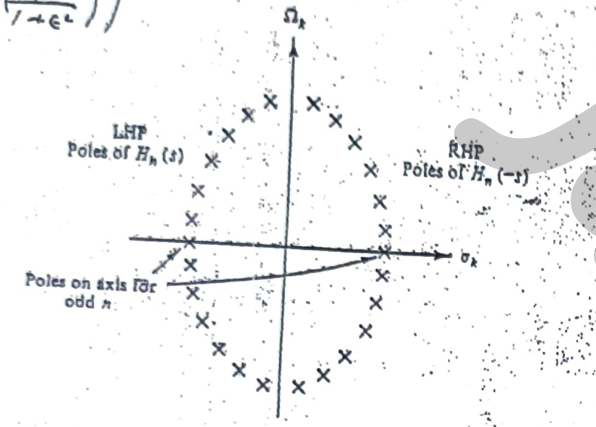
Find the transfer function $H_1(s)$ for the normalized Butterworth filter of order 1.

TABLE 3.1a BUTTERWORTH POLYNOMIALS AND NORMALIZED LOW-PASS BUTTERWORTH FILTERS

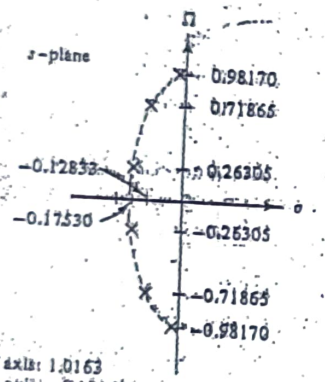
Order n	Butterworth polynomial $B_n(s)$
1	$s + 1$
2	$s^2 + \sqrt{2}s + 1$
3	$(s^2 + s + 1)(s + 1)$
4	$(s^2 + 0.7653s + 1)(s^2 + 1.8477s + 1)$
5	$(s + 1)(s^2 + 0.6180s + 1)(s^2 + 1.6180s + 1)$

Normalized low-pass Butterworth filters

$$H_n(s) = \frac{1}{B_n(s)}$$



(a)



$$\sigma_k = -a \sin\left[\frac{(2k-1)}{2n}\pi\right]$$

$$\Omega_k = b \cos\left[\frac{(2k-1)}{2n}\pi\right]$$

k	σ_k	Ω_k
1	−0.0469732	0.9817052
2	−0.1283332	0.7186581
3	−0.1753064	0.2630471
4	−0.1753064	−0.2630471
5	−0.1283332	−0.7186581
6	−0.0469732	−0.9817052

(b)

Figure 3.11 (a) Poles of $H_n(s)H_n(-s)$ for the normalized low-pass Chebyshev filter. (b) Poles of $H(s)$ for $n = 6$, $\epsilon = 0.7647831$.

Using the left half plane poles only, $H_n(s)$ can be written in the following form,

$$H_n(s) = \frac{K}{\prod_{\text{LHP poles}} (s - s_k)} = \frac{K}{V_n(s)} \tag{3.26}$$

where K is a normalizing factor whose value makes $H(0)$ equal to 1 for n odd and $1/(1 + \epsilon^2)^{1/2}$ for n even. In Eq. (3.26), $V_n(s)$ is a polynomial in s given by $V_n(s) = s^n + b_{n-1}s^{n-1} + \cdots + b_1 s + b_0$. The normalizing constant K is easily seen for n odd and even to be

$$K = V_n(0) = b_0, \qquad n \text{ odd}$$
$$K = V_n(0)/(1 + \epsilon^2)^{1/2}, \qquad n \text{ even}$$

Two different forms of tables have been established for $V_n(s)$ for various n and given values of ϵ. Table 3.4 gives the $V_n(s)$ in polynomial form for $n = 1$ to 10 and ϵ corresponding to $\frac{1}{2}$, 1, 2, and 3 dB ripples, while Table 3.5 gives the zeros [poles of $H_n(s)$] for the same n and ϵ. As for the case of the Butterworth filter, the table with the poles will be the most useful. In Fig. 3.11(b), the pole locations of $H_n(s)$ for $n = 6$ and $\epsilon = 0.7647831$ are shown. Note that the poles are not equally spaced but do exhibit symmetry with respect to the σ axis.

We see that the normalized Chebyshev low-pass filter has only two parameters, ϵ and n. Normal filter design proceeds from a set of specifications which include acceptable ripple, critical frequencies, and stopband attenuation. The acceptable ripple determines the ϵ, while the stopband attenuation and ϵ determine the order n as discussed in the following section.

Selection of n. For low-pass filter design we are usually interested not only in the cutoff frequency but in the stopband attenuation. Usual specifications might be that the magnitude squared frequency response is less than a certain value $1/A^2$ at a frequency Ω_s in the stopband as seen in Fig. 3.10. It has been shown [10] that the n that satisfies a specified ripple characterized by ϵ and a stopband gain of $1/A$ at a particular Ω_s is given by

$$n = \left\lceil \frac{\log_{10}[g + \sqrt{(g^2 - 1)^{1/2}}]}{\log_{10}[\Omega_s + \sqrt{(\Omega_s^2 - 1)^{1/2}}]} \right\rceil \tag{3.27}$$

where

$$A = 1/|H_n(j\Omega_s)|$$
$$g = [(A^2 - 1)/\epsilon^2]^{1/2} = \sqrt{\frac{A^2 - 1}{\epsilon^2}} \tag{3.28}$$

The following examples use (3.27) and (3.28) in the designs of normalized and unnormalized Chebyshev low-pass filters.

EXAMPLE 3.5

Design a low-pass 1-rad/sec bandwidth Chebyshev filter with the following characteristics:

(a) Acceptable passband ripple of 2 dB.

TABLE 3.4 POLYNOMIALS $V_n(s)$ USED IN CHEBYSHEV FILTER DESIGN FOR ¼-, 1-, 2-, and 3-dB RIPPLES.

$$\text{Chebyshev filter } H_n(s) = \frac{K_n}{V_n(s)} \qquad K_n = \begin{cases} b_0/(1+\epsilon^2)^{1/2} & \text{for } n \text{ even} \\ b_0 & \text{for } n \text{ odd} \end{cases}$$

$$V_n(s) = s^n + b_{n-1}s^{n-1} + \cdots + b_2 s^2 + b_1 s + b_0$$

n	b_0	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_9
				a. ¼ dB Ripple(ϵ = 0.3493114, ϵ^2 = 0.1220184)						
1	2.8627752									
2	1.5162026	1.4256245								
3	0.7156938	1.5348954	1.2529130							
4	0.3790506	1.0254553	1.7168662	1.1973856						
5	0.1789234	0.7525181	1.3095747	1.9373675	1.1724909					
6	0.0947626	0.4323669	1.1718613	1.5897635	2.1718446	1.1591761				
7	0.0447309	0.2820722	0.7556511	1.6479029	1.8694079	2.4126510	1.1512176			
8	0.0236907	0.1525444	0.5735604	1.1485894	2.1840154	2.1492173	2.6567498	1.1460801		
9	0.0111827	0.0941198	0.3408193	0.9836199	1.6113880	2.7814990	2.4293297	2.9027337	1.1425705	
10	0.0059227	0.0492855	0.2372688	0.6269689	1.5274307	2.1442372	3.4409268	2.7097415	3.1498757	1.140066
				b. 1-dB Ripple (ϵ = 0.5088471, ϵ^2 = 0.2589254)						
1	1.9652267									
2	1.1025103	1.0977343								
3	0.4913067	1.2384092	0.9883412							
4	0.2756276	0.7426194	1.4539248	0.9528114						
5	0.1228267	0.5805342	0.9743961	1.6888160	0.9368201					
6	0.0689069	0.3070808	0.9393461	1.2021409	1.9308256	0.9282510				
7	0.0307066	0.2136712	0.5486192	1.3575440	1.4287930	2.1760778	0.9231228			
8	0.0172267	0.1073447	0.4478257	0.8468243	1.8369024	1.6551557	2.4230264	0.9198113		
9	0.0076767	0.0706048	0.2441864	0.7863109	1.2016071	2.3781188	1.8814798	2.6709468	0.9175476	
10	0.0043067	0.0344971	0.1828512	0.4553892	1.2444914	1.6129856	2.9815094	2.1078524	2.9194657	0.915932
				c. 2-dB Ripple (ϵ = 0.7647831, ϵ^2 = 0.5848932)						
1	1.3075603									
2	0.6367681	0.8038164								
3	0.3260981	1.0221901	0.7378216							
4	0.2057651	0.5167981	1.2564819	0.7162150	0.7064606					
5	0.0817225	0.4593491	0.6934770	1.4995433	1.7458587	0.7012257				
6	0.0514433	0.2102706	0.7714618	0.8670149	1.0392203	1.9935272	0.6978929			
7	0.0204228	0.1660920	0.3825056	1.1444390	1.5795807	1.2117121	2.2422529	0.6960646		
8	0.0128603	0.0729373	0.3587043	0.5982214	1.2117121	2.0767479	1.3837464	2.4912897	0.6946793	
9	0.0051076	0.0543756	0.1684473	0.6444677	0.8568648	1.3837464	2.4912897	1.5557424	2.7406032	0.693
10	0.0032151	0.0233347	0.1440057	0.3177560	1.0389104	1.1585287	2.6362507	1.5557424	2.7406032	0.693
				d. 3-dB Ripple (ϵ = 0.9976283, ϵ^2 = 0.9952623)						
1	1.0023773									
2	0.7079478	0.6448996								
3	0.2505943	0.9283480	0.5972404							
4	0.1769869	0.4047679	1.1691176	0.5815799						
5	0.0626391	0.4079421	0.5488626	1.4149847	0.5744296					
6	0.0442467	0.1634299	0.6990977	0.6906098	1.6628481	0.5706979				
7	0.0156621	0.1461530	0.3000167	1.0518448	0.8314411	1.9115507	0.5684201			
8	0.0110617	0.0564813	0.3207646	0.4718990	1.4666990	0.9719473	2.1607148	0.5669476		
9	0.0039154	0.0475900	0.1313851	0.5834984	0.6789075	1.9438443	1.1122863	2.4101346	0.5659234	
10	0.0027654	0.0180313	0.1277560	0.2492043	0.9499208	0.9210659	2.4834205	1.2526467	2.6597378	0.565

Source: Weinberg, Louis. *Network Analysis and Synthesis*. McGraw-Hill, New York, 1962. Reprinted with permission of the author.

(b) Cutoff radian frequency of 1 rad/sec.
(c) Stopband attenuation of 20 dB or greater beyond 1.3 rad/sec.

Solution. To satisfy the ripple and stopband attenuation requirements, we see from the magnitude squared plot of Fig. 3.10 that

$$20 \log |H_n(j1)| = 20 \log [1/(1 + \epsilon^2)]^{1/2} = 10 \log [1/(1 + \epsilon^2)] = -2$$
$$20 \log |H_n(j1.3)| = 20 \log |(1/A^2)^{1/2}| = 20 \log (1/A) = -20$$

The above equations are easily solved giving an $\epsilon = 0.76478$ and an $A = 10$. Using (3.27) and (3.28), the corresponding g and n are determined as follows:

$$g = [(100 - 1)/(0.76478)^2]^{1/2} = 13.01$$
$$n = \left\lceil \frac{\log_{10}\{13.01 + [(13.01)^2 - 1]^{1/2}\}}{\log_{10}\{1.3 + [(1.3)^2 - 1]^{1/2}\}} \right\rceil = \lceil 4.3 \rceil = 5$$

Using the 2-dB ripple part of Table 3.3 for $n = 5$, and the fact that $H(0) = 1$ since n is odd, we have the desired Chebyshev unit bandwidth low-pass filter as

$$H_5(s) = K/(s^5 + b_4 s^4 + \cdots + b_1 s + b_0)$$
$$= \frac{0.08172}{s^5 + 0.70646s^4 + 1.4995s^3 + 0.6934s^2 + 0.459349s + 0.08172}$$

An expression for $H_5(s)$ in terms of quadratic factors obtained by using the poles from Table 3.4 is as follows,

$$H_5(s) = \frac{0.08172}{(s + 0.218303)[s - (-0.06746 + 0.97345j)][s - (-0.06746 - 0.97345j)]} \times \frac{1}{[s - (-0.1766151 + 0.6016j)][s - (-0.1766151 - 0.6016j)]}$$

This type of quadratic form proves useful in realizing digital filters by cascading as will be shown in Chapter 4. Multiplying out the pairs of linear factors with complex conjugate pairs results in

$$H_5(s) = \frac{0.08172}{}$$

EXAMPLE 3.6

Design a Chebyshev low-pass filter to satisfy the following specifications: (a) acceptable passband ripple of 2 dB, (b) cutoff frequency of 40 rad/sec, and (c) stopband attenuation of 20 dB or more at 52 rad/sec.

Solution. The general approach is to first change the requirements to those of a low-pass unit bandwidth prototype, design such a low-pass filter, and then apply a low-pass to low-pass transformation to that prototype.

From Table 3.2 we see that if s is replaced by s/Ω_u, a 1-rad/sec LPF with critical frequency Ω_r is transformed to an LPF with cutoff Ω_u and a critical frequency Ω_r'. The required Ω_r for the normalized filter is obtained from the backward design equation of Table 3.2 as

$$\Omega_r = \Omega_r'/\Omega_u = 52/40 = 1.3$$

Therefore we need to design a normalized low-pass Chevyshev filter with 2-dB cutoff at 1-rad/sec and a 20-dB attenuation at 1.3 rad/sec and then apply the transformation $s \to s/40$. As it happens, just such a low-pass filter was designed in Example 3.5. So, by using the $H_5(s)$ of Example 3.5, $H_d(s)$ can be written as

$$H_d(s) = H_5(s)\big|_{s \to s/40}$$
$$= \frac{0.08172}{(s/40 + 0.218303)(s^2/40^2 + 0.134922s/40 + 0.95215)} \times \frac{1}{s^2/40^2 + 0.35323s/40 + 0.393115)}$$
$$= \frac{8.366 \times 10^6}{(s + 8.73212)(s^2 + 5.3969s + 1523.44)(s^2 + 14.1292s + 628.984)}$$

The designs of Chebyshev high-pass, bandpass, and band-elimination filters are easily accomplished in a parallel manner. First, the backward design formulas of Table 3.2 are used to obtain low-pass filter requirements; second, a Chebyshev filter is found to satisfy those requirements; and third, a transformation is performed to get the required $H_d(s)$.

It is worth noting at this time that after the $H_d(s)$ has been found for the analog filter, a synthesis procedure must be undertaken to obtain the implementation in physical components, i.e., resistors, capacitors, inductors, etc.

TABLE 3.5 ZEROS OF POLYNOMIAL $V_n(s)$ DERIVED FROM THE CHEBYSHEV APPROXIMATION FOR ½-, 1-, 2-, and 3-dB RIPPLES

Chebyshev filter $H_n(s) = \dfrac{K_n}{V_n(s)}$, $K_n = \begin{cases} b_0 \\ b_0/(1+\epsilon^2)^{1/2} \end{cases}$ for n odd, for n even

$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$	$n = 9$	$n = 10$
a. ½-dB Ripple ($\epsilon = 0.3493114$, $\epsilon^2 = 0.1220184$)									
−2.8627752	−0.7128122 ± j1.0040425	−0.6264565	−0.1755531 ± j1.0162529	−0.3623196	−0.0776501 ± j1.0084608	−0.2561700	−0.0436201 ± j1.0050021	−0.1984053	−0.02789 ± j1.00327
		−0.3132282 ± j1.0219275	−0.4233398 ± j0.4209457	−0.1119629 ± j1.0113574	−0.2121440 ± j0.7382446	−0.0570032 ± j1.0064085	−0.1242195 ± j0.8519996	−0.0344527 ± j1.0040040	−0.08096 ± j0.90506
				−0.2931227 ± j0.6251768	−0.2897940 ± j0.2702162	−0.1597194 ± j0.8070770	−0.1859076 ± j0.5692879	−0.0992026 ± j0.8829063	−0.12610 ± j0.71826
						−0.2308012 ± j0.4478939	−0.2192929 ± j0.1999073	−0.1519873 ± j0.6553170	−0.15890 ± j0.46115
								−0.1864400 ± j0.3486869	−0.17614 ± j0.15890
b. 1-dB Ripple ($\epsilon = 0.5088471$, $\epsilon^2 = 0.2589254$)									
−1.9652267	−0.5488672 ± j0.8951286	−0.4941706	−0.1395360 ± j0.98333792	−0.2894933	−0.0621810 ± j0.9934115	−0.2054141	−0.0350082 ± j0.9964513	−0.1593305	−0.02241 ± j0.99772
		−0.2470853 ± j0.9659987	−0.3368697 ± j0.4073290	−0.0894584 ± j0.9901071	−0.1698817 ± j0.7272275	−0.0457089 ± j0.9952839	−0.0996950 ± j0.8447506	−0.0276674 ± j0.9922297	−0.10131 ± j0.71432
				−0.2342050 ± j0.6119198	−0.2320627 ± j0.2661837	−0.1280736 ± j0.7981557	−0.1492041 ± j0.5644443	−0.0796652 ± j0.8769490	−0.06504 ± j0.90010
						−0.1850717 ± j0.4429430	−0.1759983 ± j0.1982065	−0.1220542 ± j0.6508954	−0.12766 ± j0.45862
								−0.1497217 ± j0.3463342	−0.14151 ± j0.15803
c. 2-dB Ripple ($\epsilon = 0.7647831$, $\epsilon^2 = 0.5848932$)									
−1.3075603	−0.4019082 ± j0.6031750	−0.3689108	−0.1048872 ± j0.9579530	−0.2163683	−0.0469326 ± j0.9817054	−0.1552958	−0.0261924 ± j0.9893870	−0.1206298	−0.0169953 ± j0.9934868
		−0.1844354 ± j0.9230771	−0.2532202 ± j0.3967971	−0.0674610 ± j0.9734557	−0.1283332 ± j0.7186581	−0.0345566 ± j0.9866139	−0.0754439 ± j0.8391009	−0.0209471 ± j0.9919471	−0.0767332 ± j0.7112580
				−0.1766151 ± j0.6016287	−0.1753064 ± j0.2630471	−0.0968253 ± j0.7912029	−0.1129098 ± j0.5606693	−0.0603149 ± j0.8723036	−0.0492657 ± j0.8962374
						−0.1399167 ± j0.4390845	−0.1331862 ± j0.1968809	−0.0924078 ± j0.6474475	−0.0966894 ± j0.4566558
								−0.1133549 ± j0.3444996	−0.1071810 ± j0.1573528
d. 3-dB Ripple ($\epsilon = 0.9976283$, $\epsilon^2 = 0.9952623$)									
1.0023773	−0.3224498 ± j0.7771576	−0.2986202	−0.0851704 ± j0.9464844	−0.1775085	−0.0382295 ± j0.9764060	−0.1264854	−0.0215782 ± j0.9867664	−0.0982716	−0.0138320 ± j0.9915418
		−0.1493101 ± j0.9038144	−0.2056195 ± j0.3920467	−0.0548531 ± j0.9659238	−0.1044450 ± j0.7147788	−0.0281456 ± j0.9826957	−0.0614494 ± j0.8365401	−0.0170647 ± j0.9895515	−0.0401419 ± j0.8944827
				−0.1436074 ± j0.5969738	−0.1426745 ± j0.2616272	−0.0788623 ± j0.7880608	−0.0919655 ± j0.5589582	−0.0491358 ± j0.8701971	−0.0625225 ± j0.7098655
						−0.1139594 ± j0.4373407	−0.1084807 ± j0.1962800	−0.0752804 ± j0.6458839	−0.0787829 ± j0.4557617
								−0.092345 ± j0.3436677	−0.0873316 ± j0.1570448

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